

Multimodal NLP and Agentic AI

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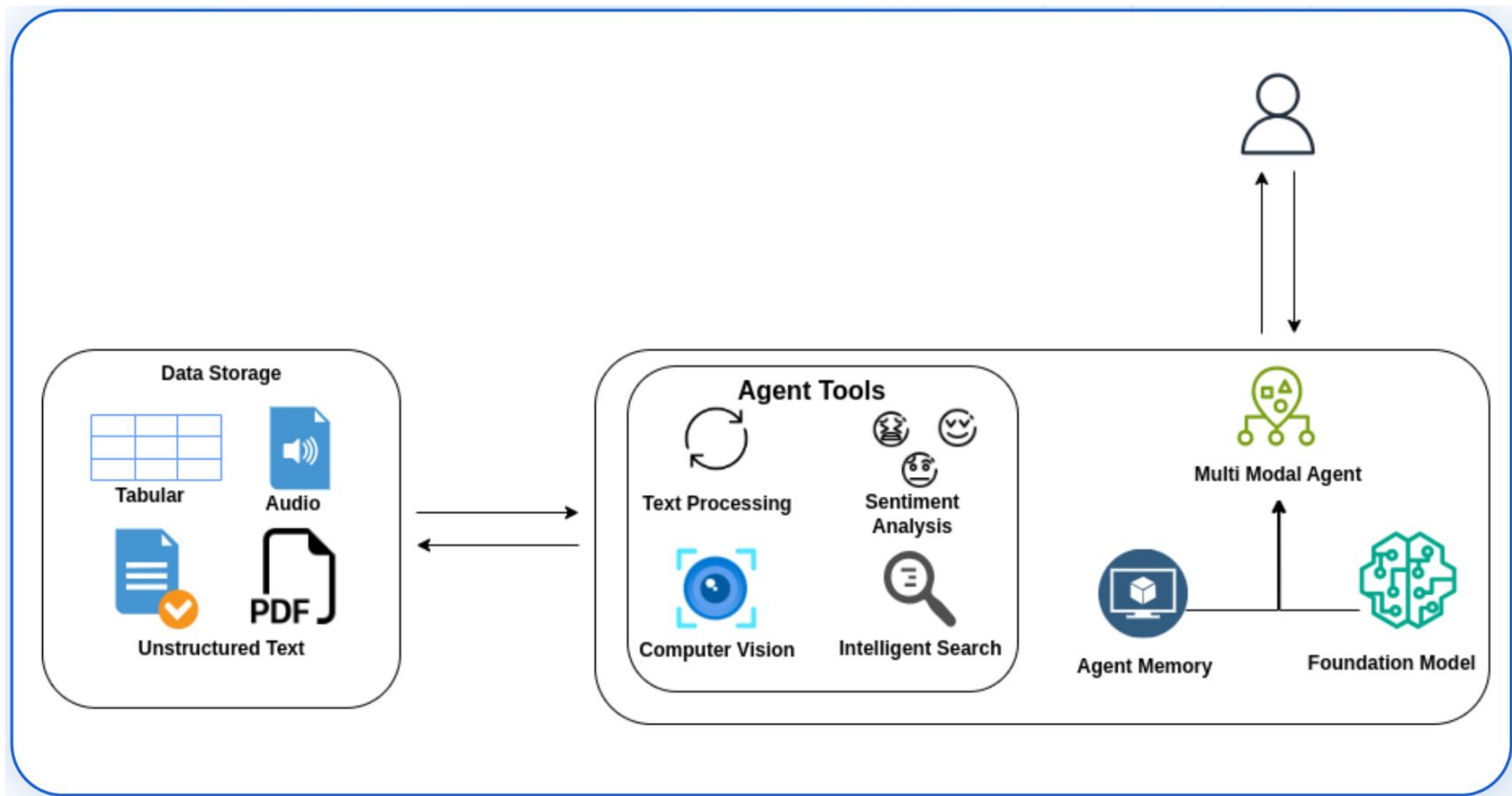


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- ▶ Traditional NLP models are **unimodal** (text-only) and **reactive**.
- ▶ Modern AI agents need to perceive, reason, and act in complex environments using **multiple modalities** (e.g., vision, text, speech).
- ▶ **Multimodal systems** and **agentic AI** aim to build goal-directed, autonomous, and continuously learning agents.
- ▶ Rise of agent frameworks (Auto-GPT, BabyAGI) and vision-language models enables new research frontiers in decision-making, robotics, and assistive AI.

What is multimodality?

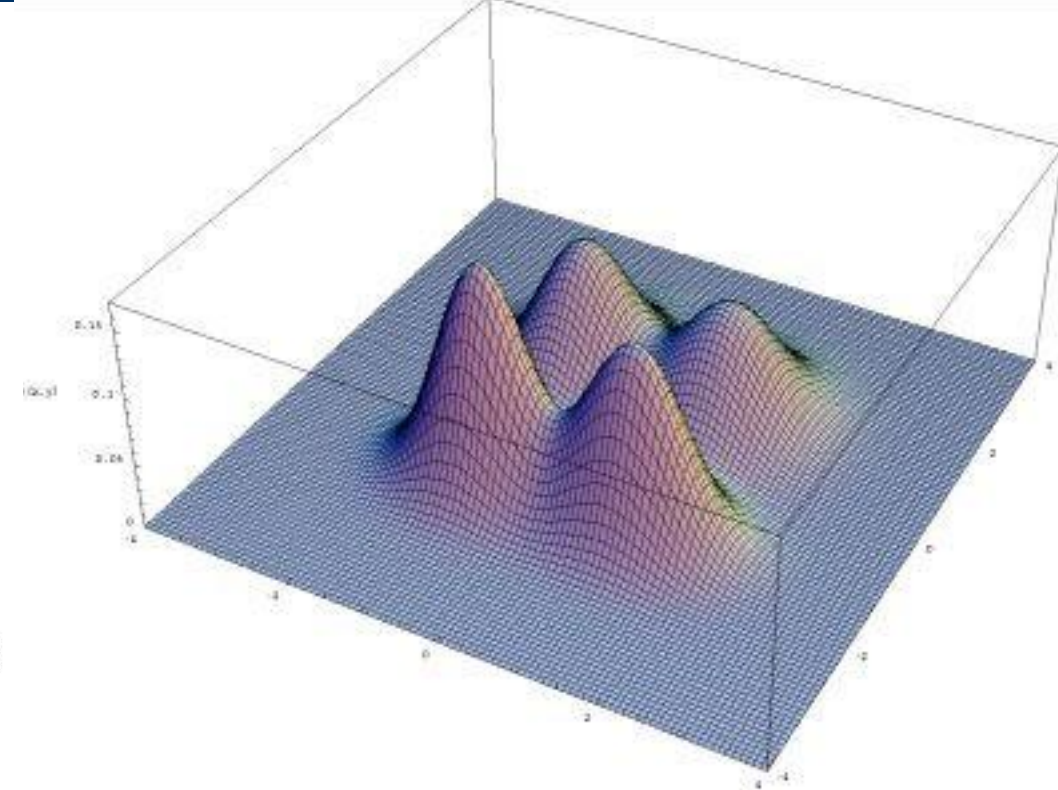
multimodal adjective

mul·ti·mod·al (ˌməl-tē-ˈmō-dəl ) -tī-

: having or involving several modes, modalities, or maxima

| *multimodal* distributions

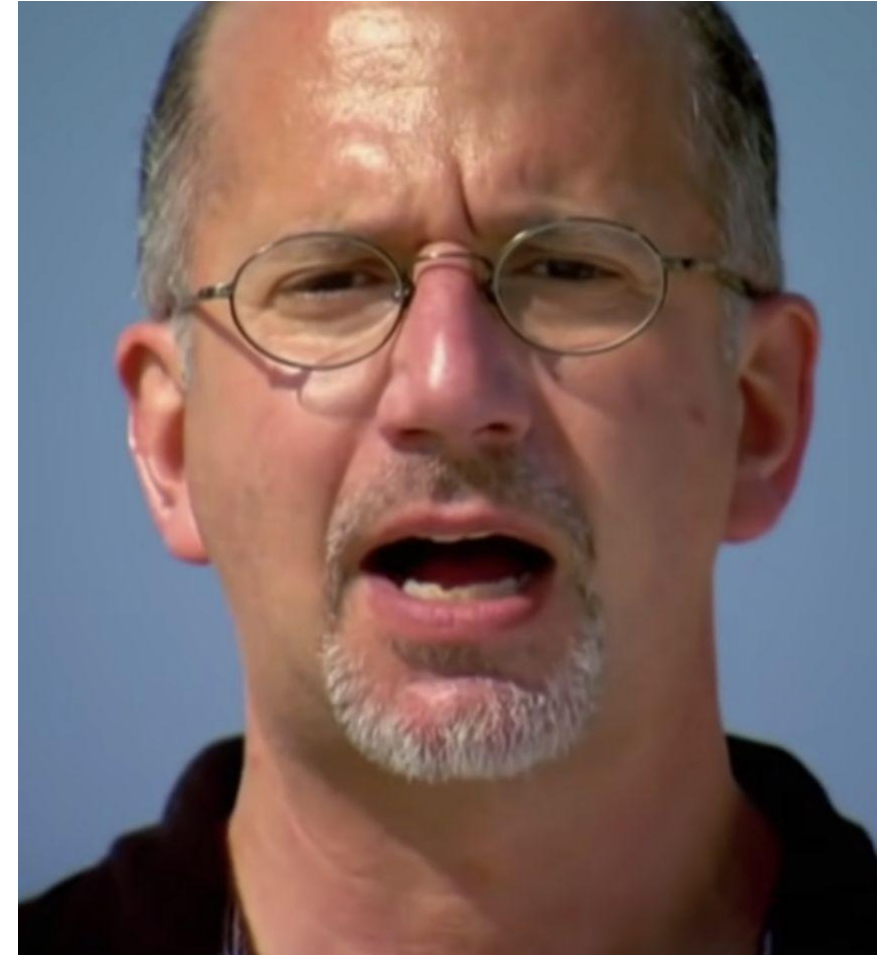
| *multimodal* therapy



In our case, focusing on NLP: text + one or more other *modality* (images, speech, audio, olfaction, others). We'll mostly focus on images as the other modality.

McGurk effect (McGurk & MacDonald, 1976)

<https://www.youtube.com/watch?v=2k8fHR9jKVM>



- ▶ Explain core vision-language models (e.g., CLIP, VisualBERT, FLAVA).
- ▶ Compare multimodal architectures and their integration mechanisms.
- ▶ Understand the principles of agentic AI and agentic loops.
- ▶ Differentiate between reactive and proactive agent behavior.
- ▶ Describe continual learning challenges in agentic systems.
- ▶ Critically evaluate frameworks like Auto-GPT and BabyAGI.
- ▶ Reflect on safety, control, and ethical issues in agentic AI.

Vision-Language Models



- ▶ Developed by OpenAI (Radford et al., 2021)
- ▶ Trained on 400M (image, text) pairs
- ▶ Learns a joint embedding space using contrastive loss
- ▶ Image and text encoders are trained to match correct pairs

Use cases:

- ▶ Zero-shot classification
- ▶ Semantic similarity
- ▶ Prompt-based image querying



Architecture:

- ▶ ResNet / ViT for images
- ▶ Transformer for text
- ▶ Cosine similarity loss

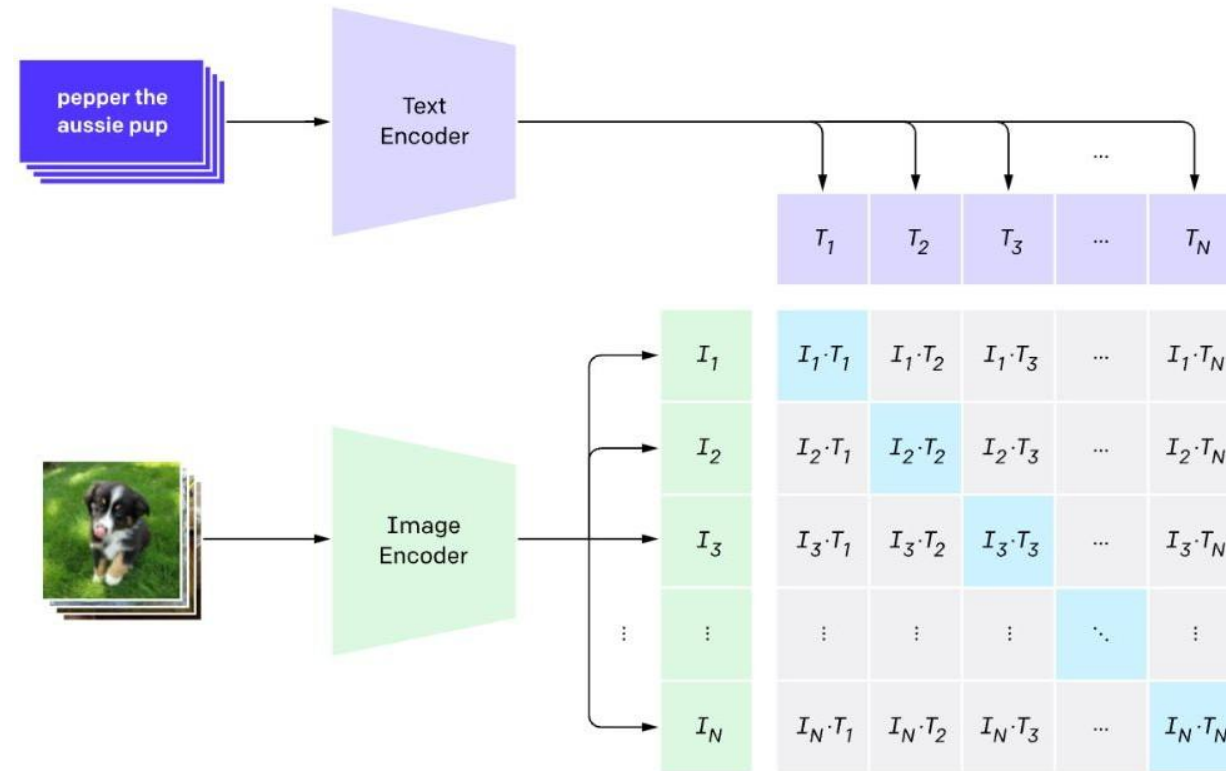
Paper: CLIP: Learning Transferable Visual Models From Natural Language Supervision

<https://arxiv.org/abs/2103.00020>

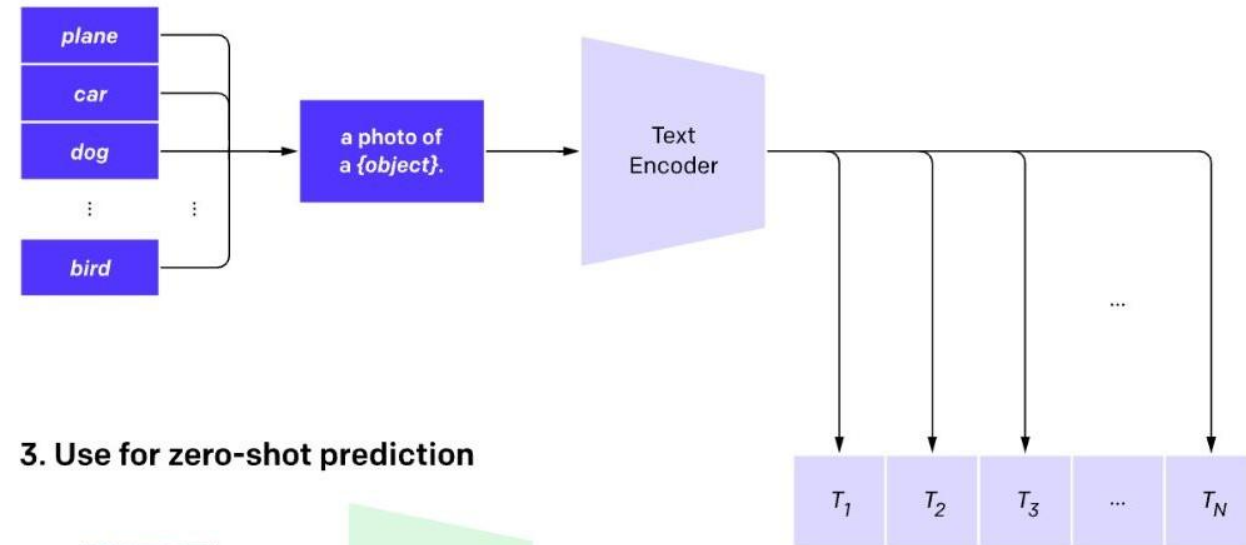
CLIP (Radford et al. 2021)

Exact same contrastive loss as earlier, but.. Transformers and *web data*!

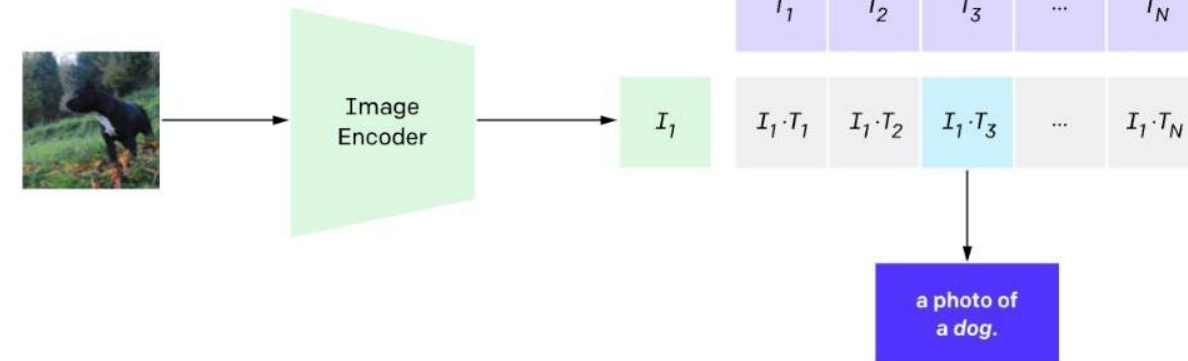
1. Contrastive pre-training



2. Create dataset classifier from label text

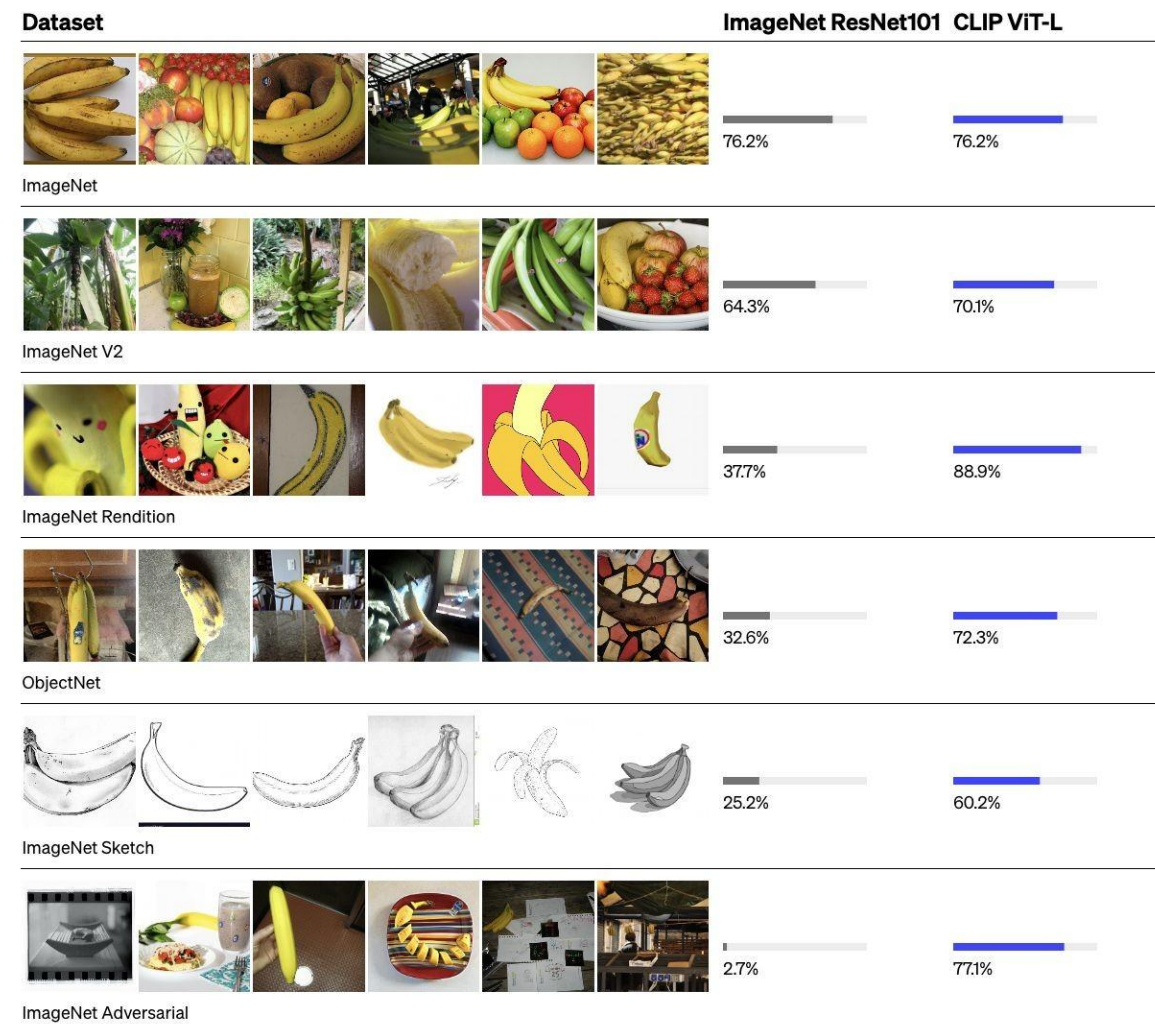


3. Use for zero-shot prediction



IMHO one of the best papers ever written in our field: extremely thorough, worth a close read.

Generalizes MUCH better →



- ▶ Joint Transformer-based model: processes both image regions and text.
- ▶ **Early fusion:** image embeddings are added as input tokens to the Transformer.
- ▶ **Pretraining objectives:**
 - Masked Language Modeling (MLM)
 - Next Sentence Prediction (NSP)
 - Image-Text Alignment

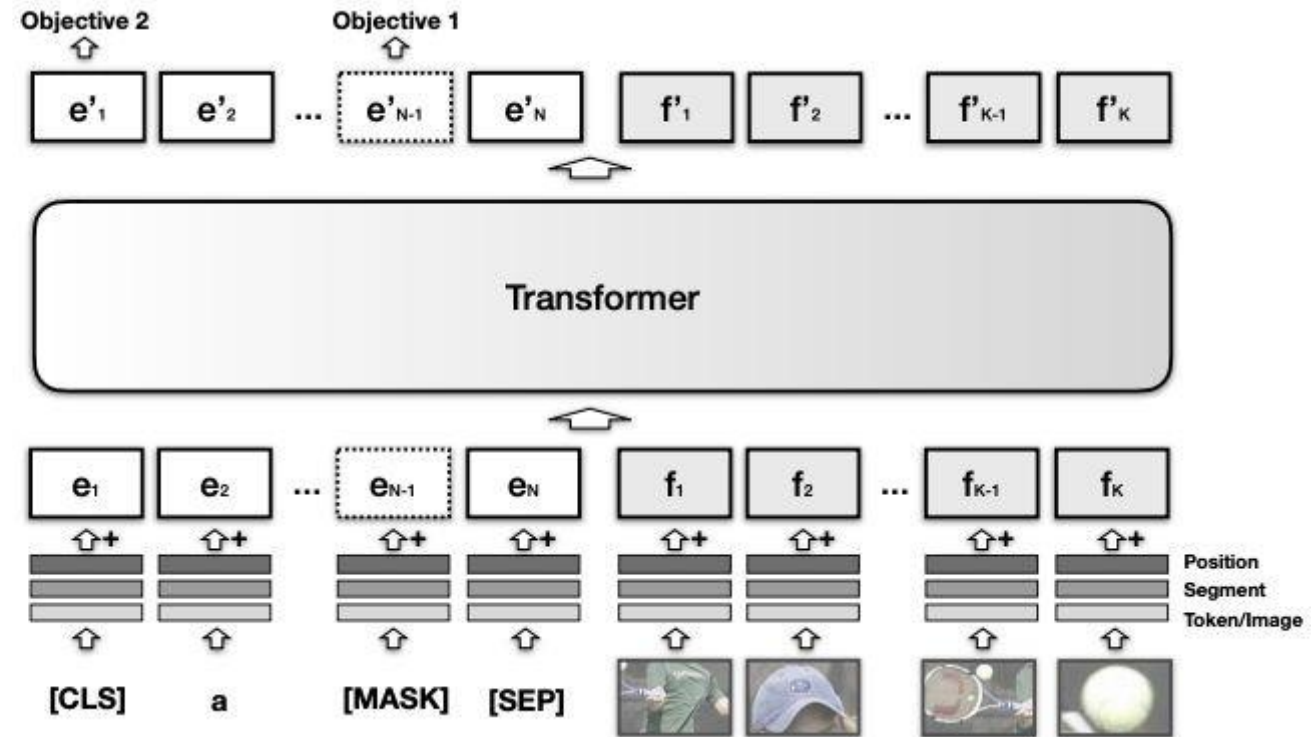
Applications:

- ▶ Visual Question Answering (VQA)
- ▶ Image captioning
- ▶ Visual commonsense reasoning

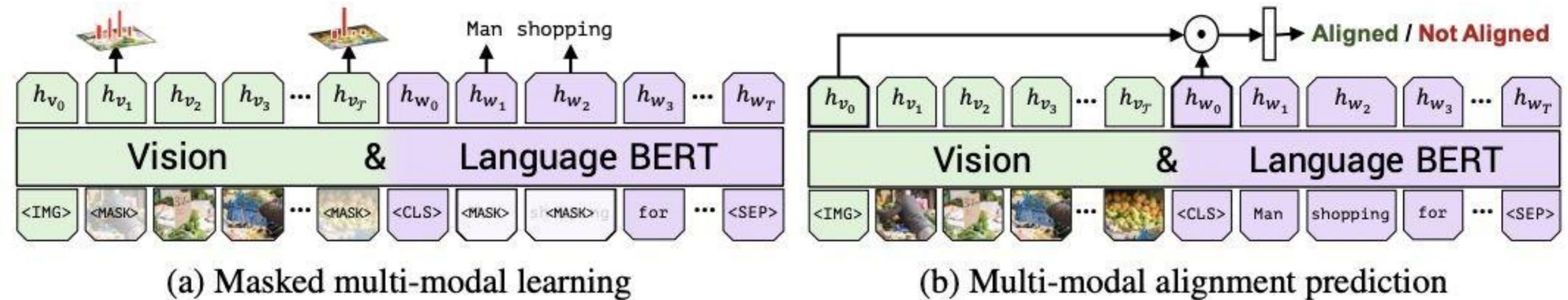
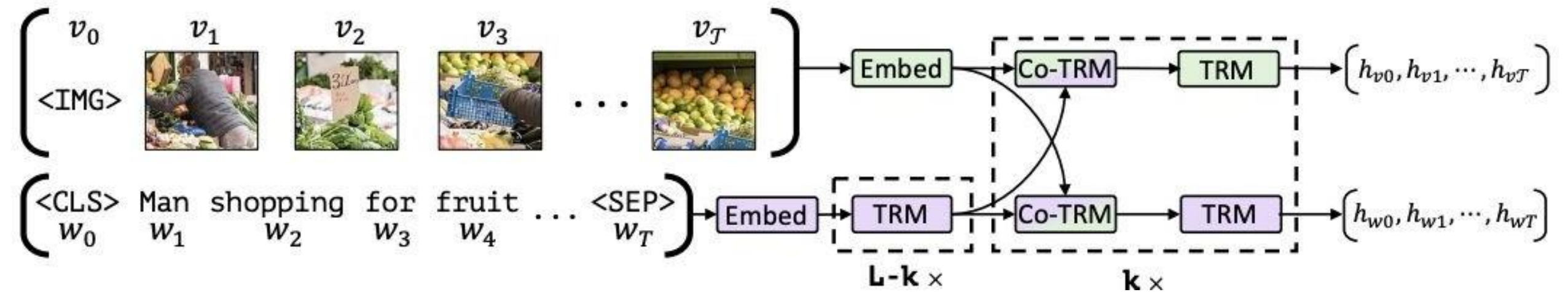
Paper: VisualBERT: A Simple and Performant Baseline for Vision and Language



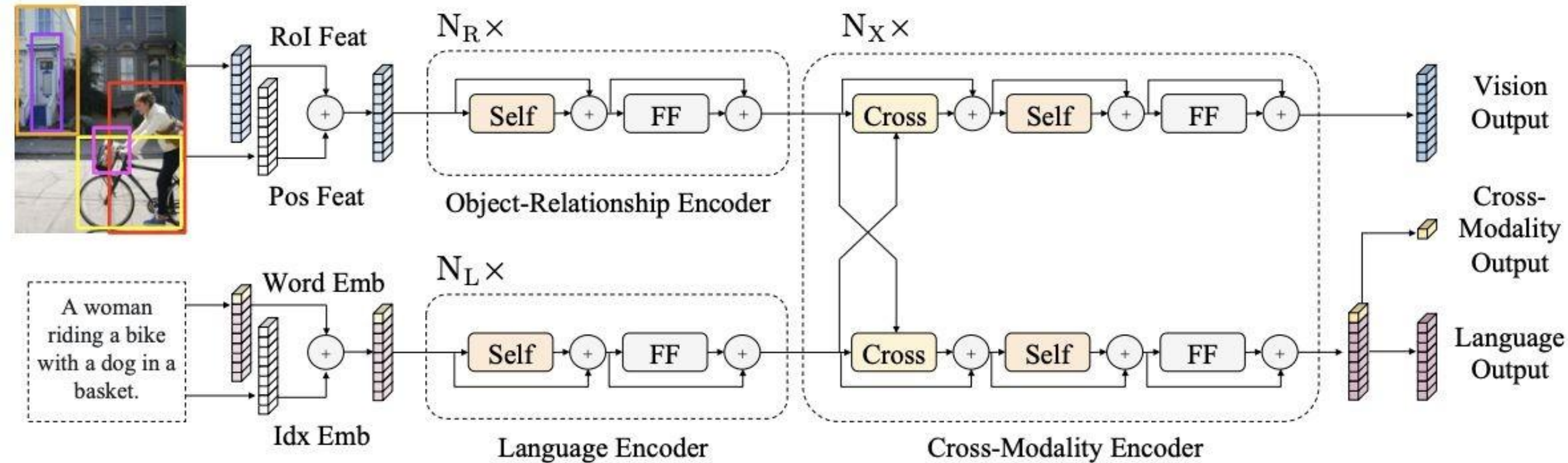
A person hits a ball with a tennis racket



VisualBERT Li et al. 2019



Learning Cross-Modality Encoder Representations from Transformers



LXMERT Tan & Bansal 2019

Visual BERTs: Supervised Multimodal Bitransformers

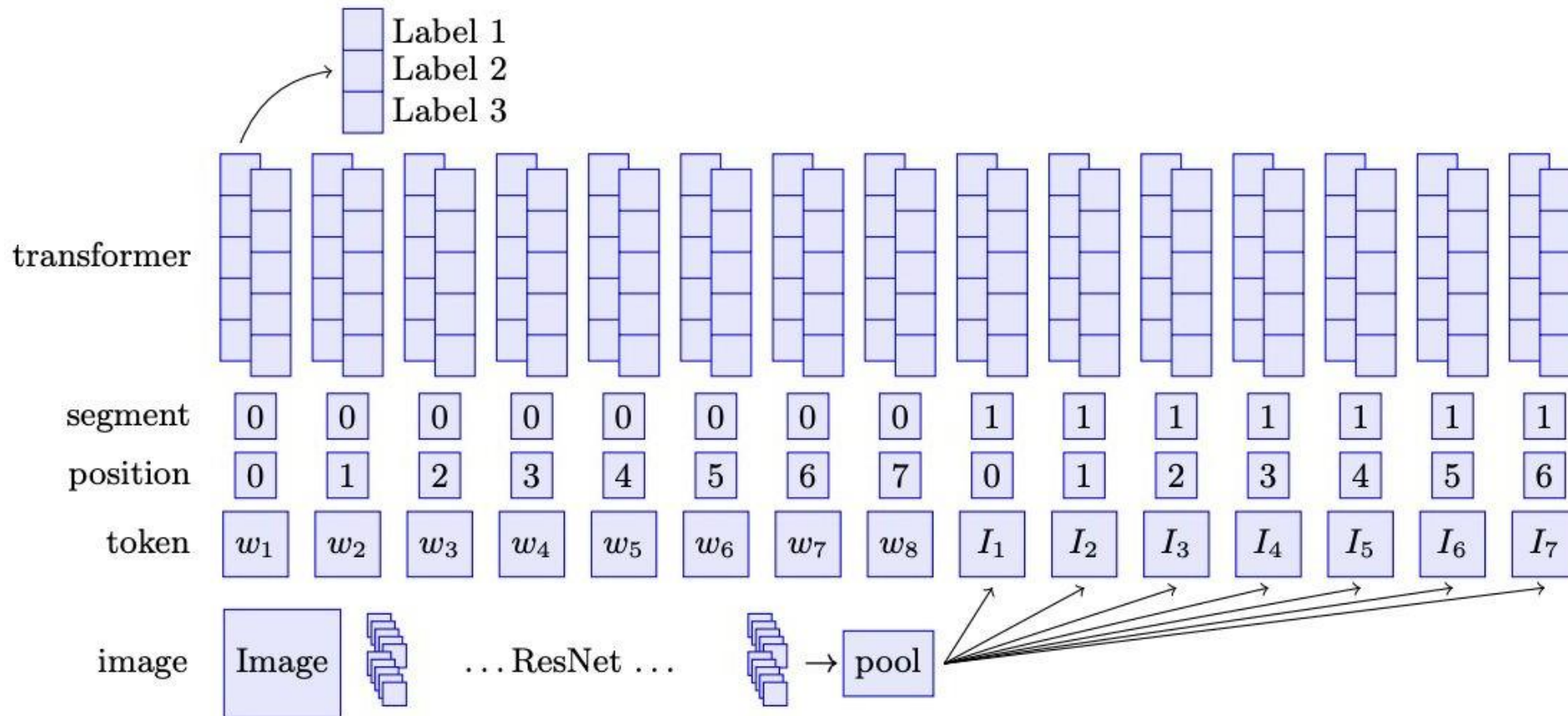
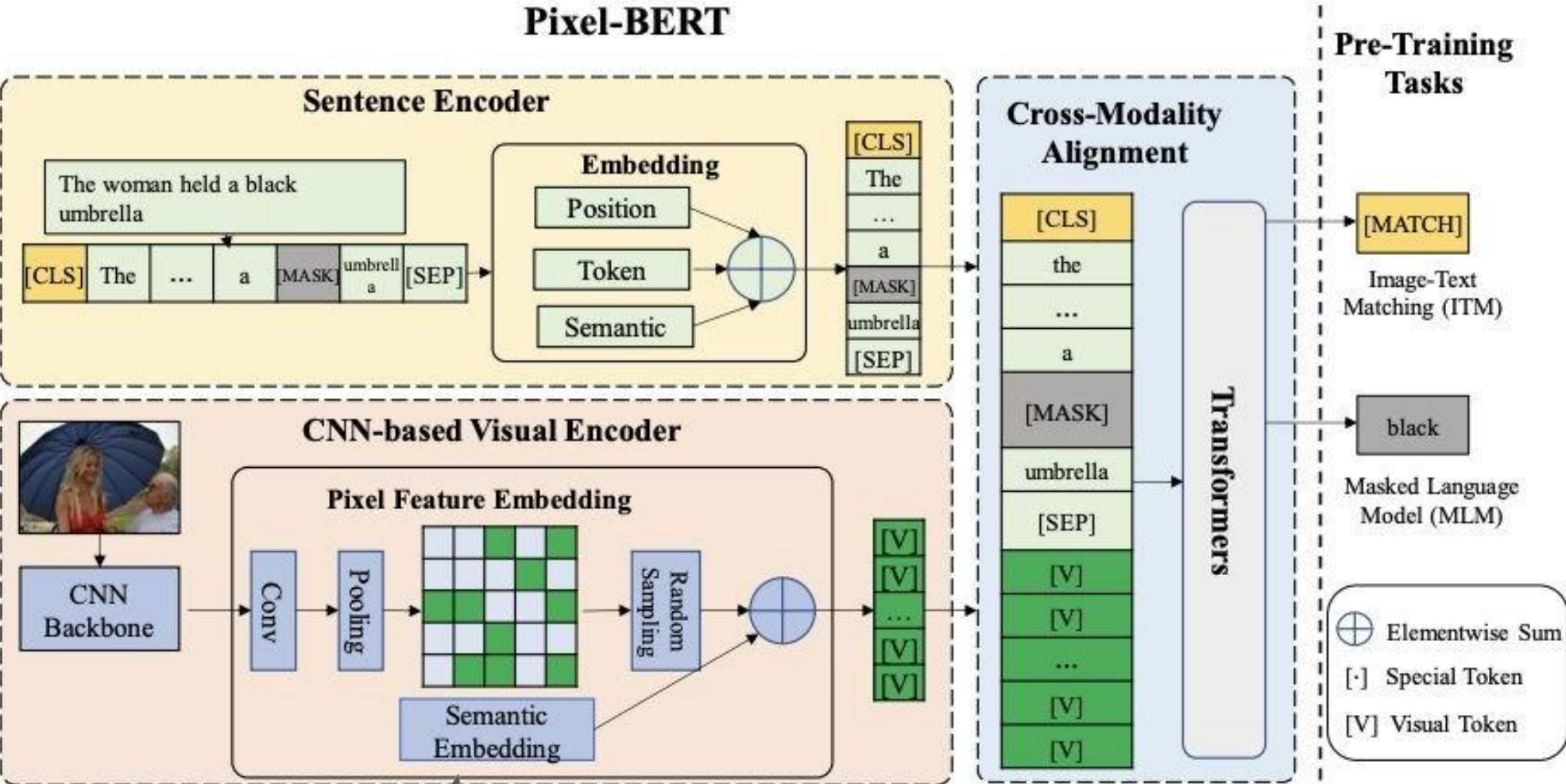


Figure 1: Illustration of the multimodal bitransformer architecture.

MMBT Kiela et al. 2019



Misnomer: they mean segment embedding

PixelBert Huang et al. 2020

- ▶ FLAVA = **Fusion of Language And Vision Architecture**
- ▶ Handles both unimodal and multimodal tasks
- ▶ **Three towers:** vision, language, multimodal
- ▶ Self-supervised objectives: masked modeling, contrastive learning

Capabilities:

- ▶ Text classification
- ▶ Image classification
- ▶ Visual Question Answering (VQA)
- ▶ Image captioning

Paper: FLAVA: A Foundational Language And Vision Alignment Model

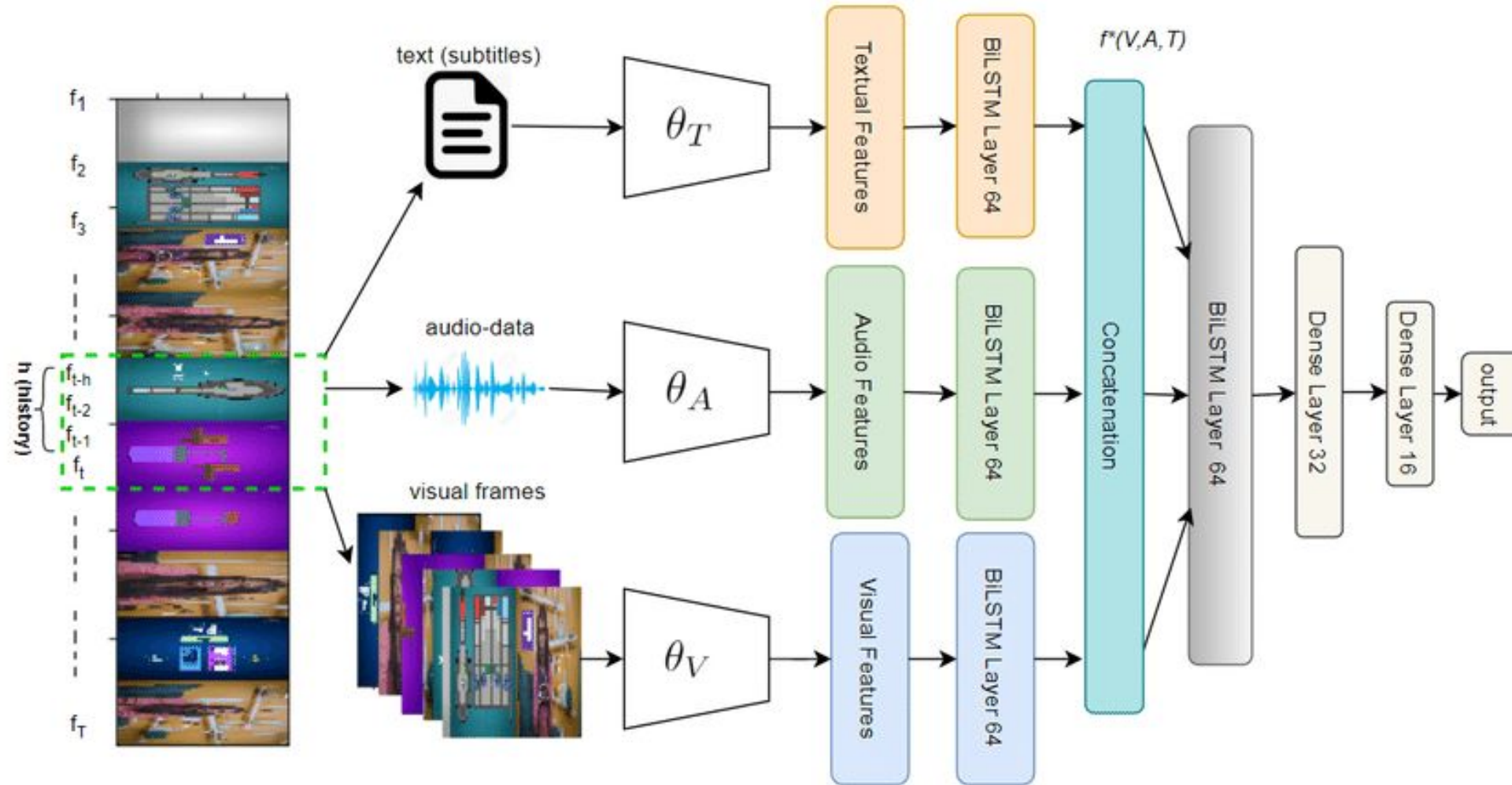
Holistic approach to multimodality.
One foundation model spanning V&L, CV and
NLP. Jointly pre-trained on:

- unimodal text data (CCNews + BookCorpus)
- unimodal image data (ImageNet)
- public paired image-text data (70M)

All data/models are publicly released.



Multimodal Architectures

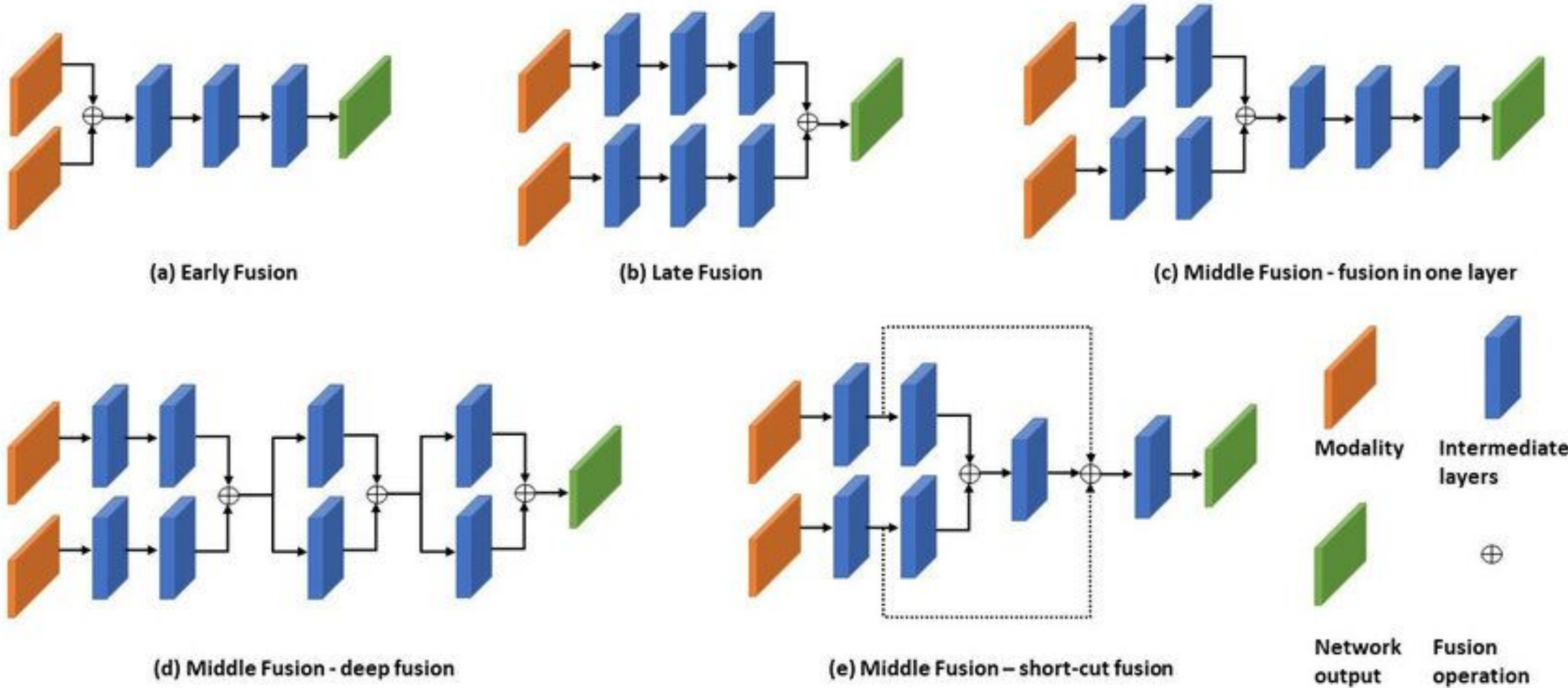


- ▶ **Early Fusion:** Combine raw modalities at the input level.
Example: VisualBERT adds image embeddings as tokens to the Transformer.
- ▶ **Late Fusion:** Process each modality separately, then merge representations.
Example: CLIP encodes images and text independently, then compares embeddings.
- ▶ **Hybrid / Cross-modal Attention:** Enable interactions between modalities at multiple stages using attention mechanisms.

Design Considerations:

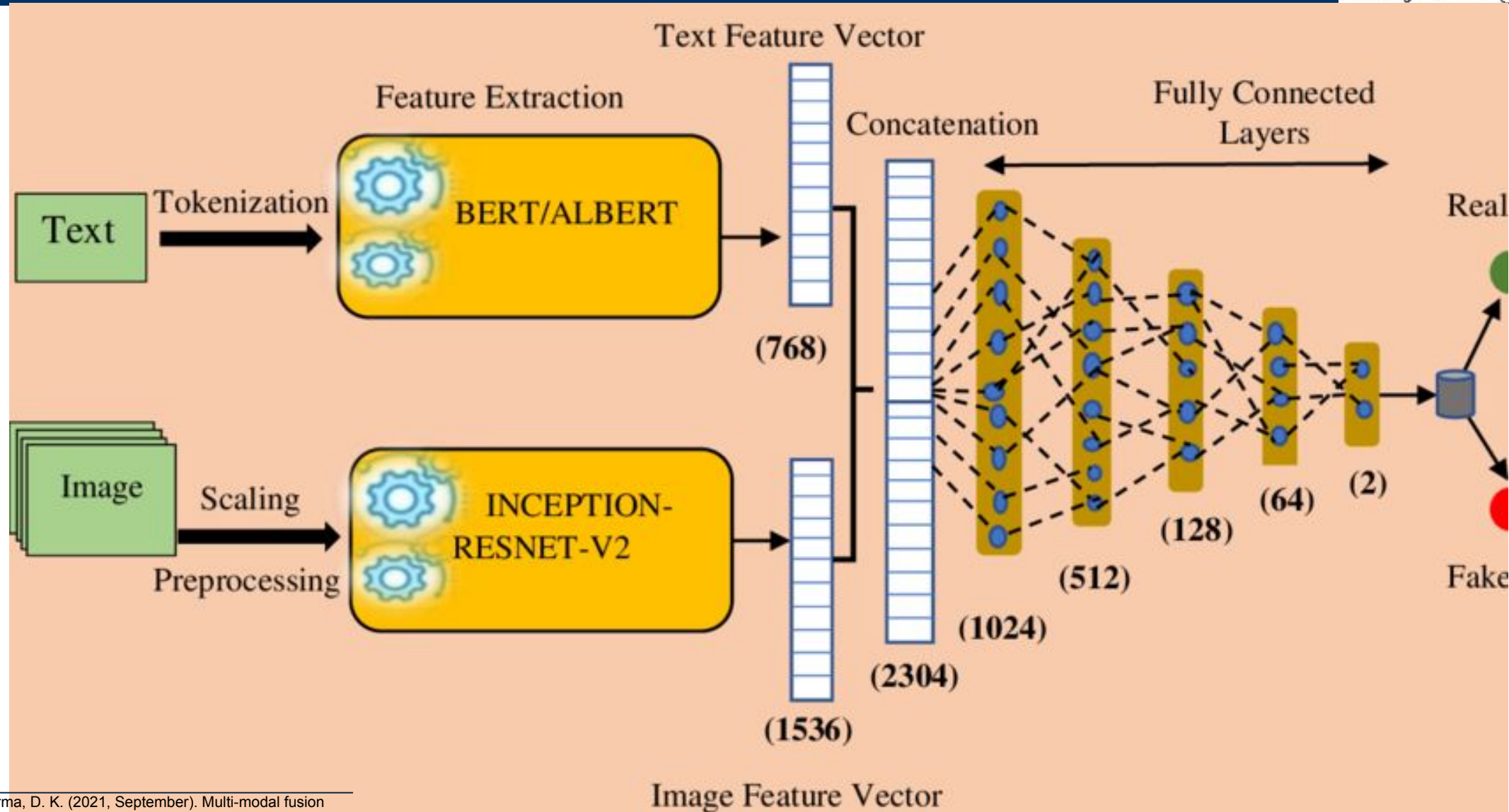
- ▶ **Alignment:** How to align representations across modalities.
- ▶ **Modality Dropout:** Handling missing or noisy modalities during training or inference.
- ▶ **Shared vs. Modality-Specific Layers:** Deciding which layers are shared and which are unique to each modality.

Categories of Multimodal Architectures

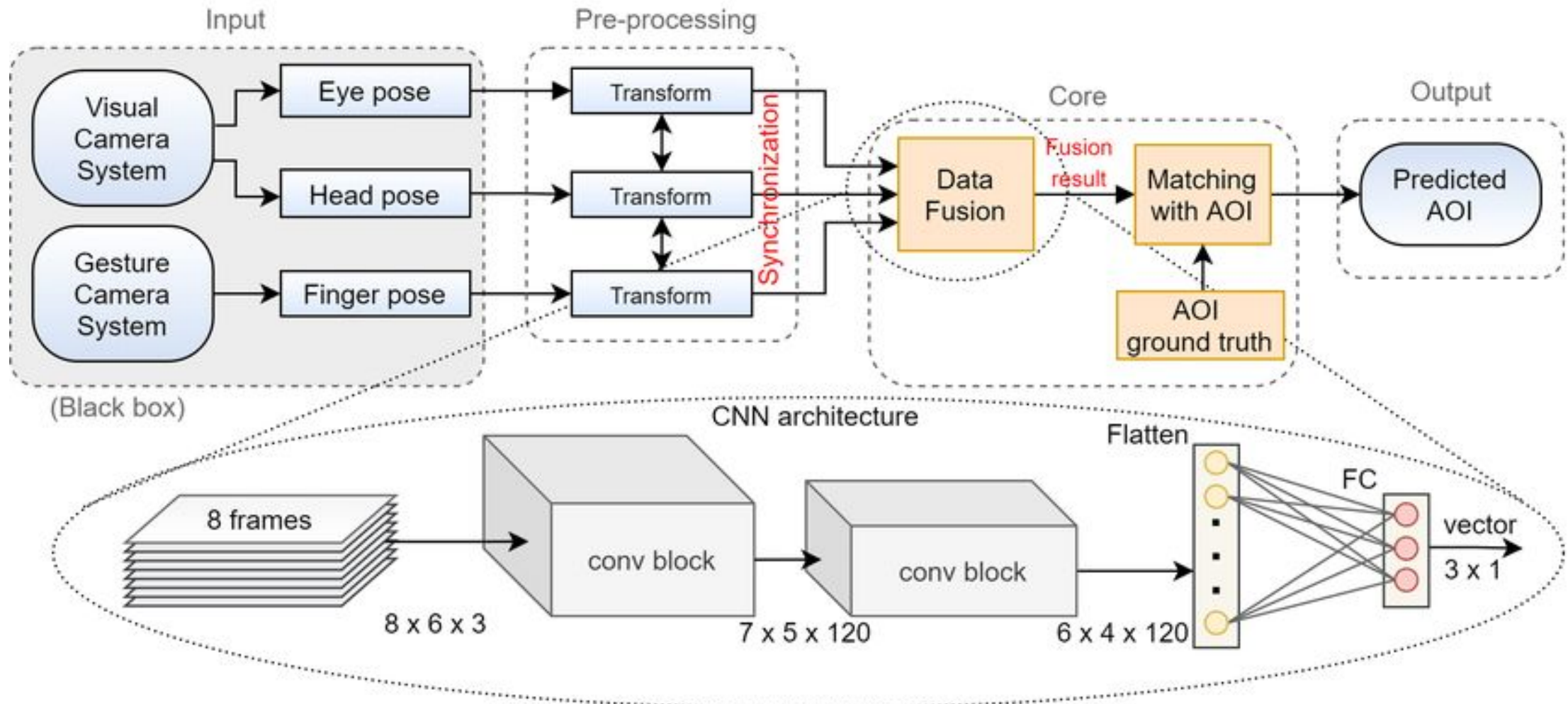


Buckley, M. (2024, September 25). Understanding and Attacking Meta's Chameleon Model.

Early Fusion



Late Fusion



Challenges in Multimodal Integration

MultiModal Learning Challenges



Representation

Finding a good representation is crucial for the success of multimodal models that combine different modalities such as image, text, and audio.



Fusion

Fusing information from different modalities to perform a prediction task is the core of multimodal learning and can be challenging due to the diverse nature of multimodal data.



Alignment

Aligning data from different modalities to create modality-invariant representations is necessary to combine information effectively.



Translation

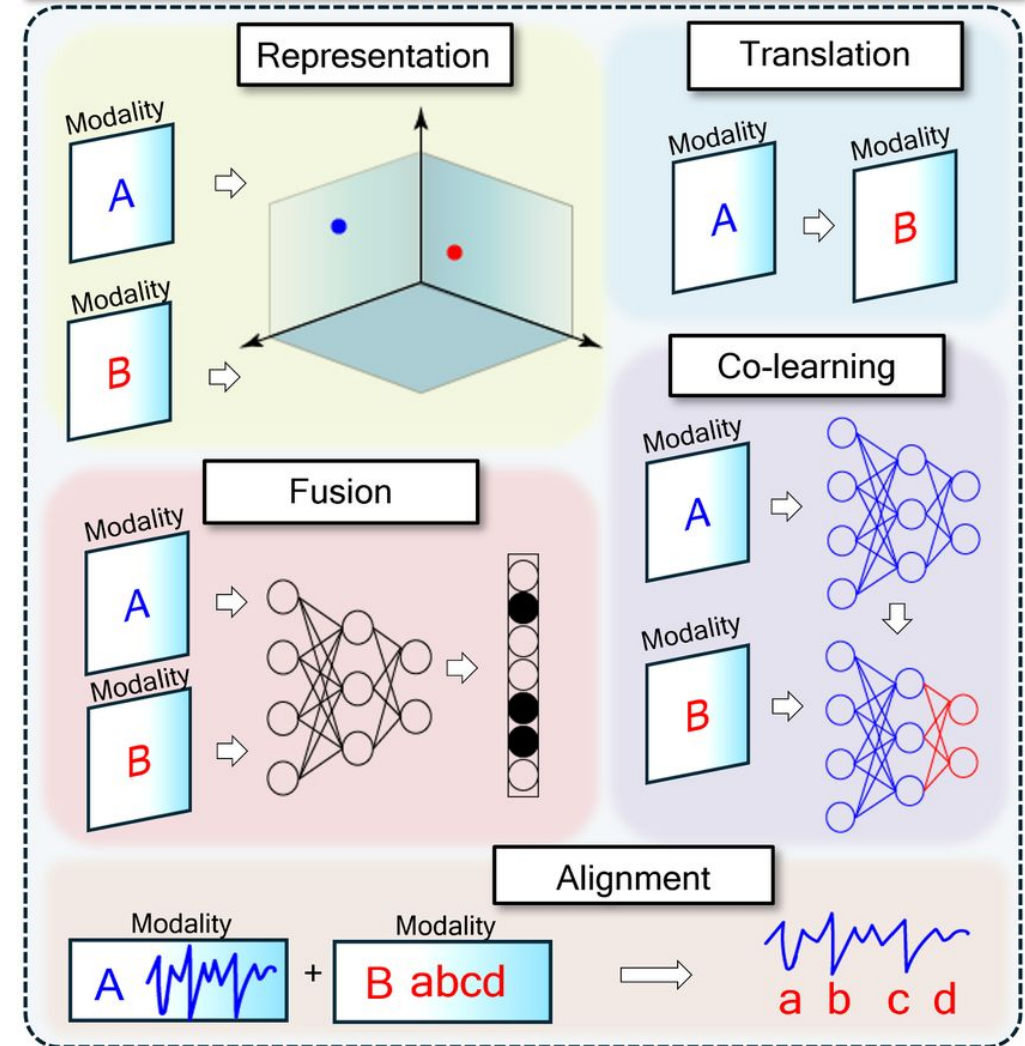
Translating data from one modality to another is a crucial task in multimodal learning, but can be difficult due to the subjective and open-ended nature of translations.



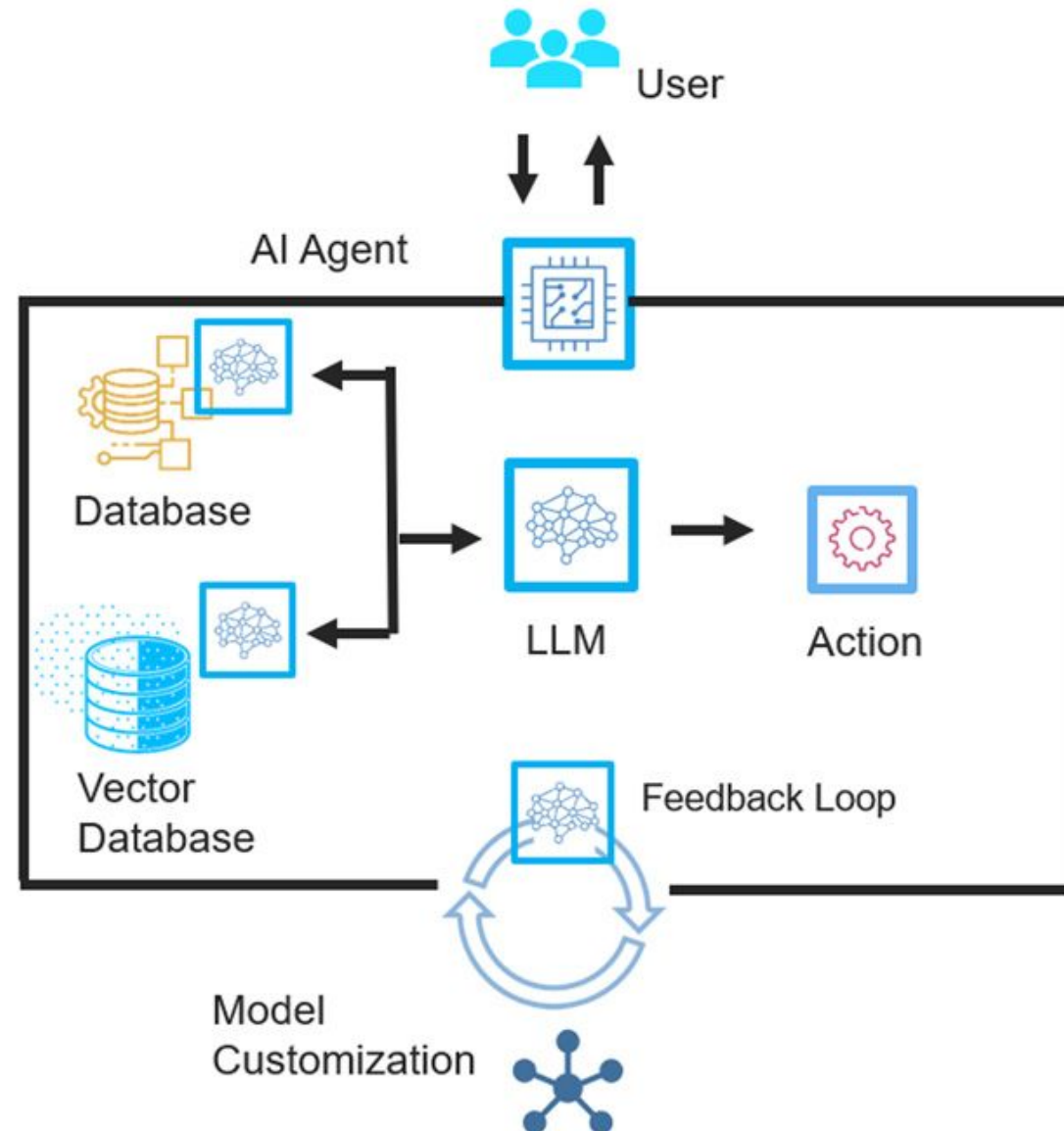
Co-learning

Co-learning aims to transfer knowledge learned through one or more modalities to tasks involving another. It is useful in medical diagnosis combining CT and MRI scans.

Multimodal Machine Learning Challenges



Agentic AI



What is Agentic AI?

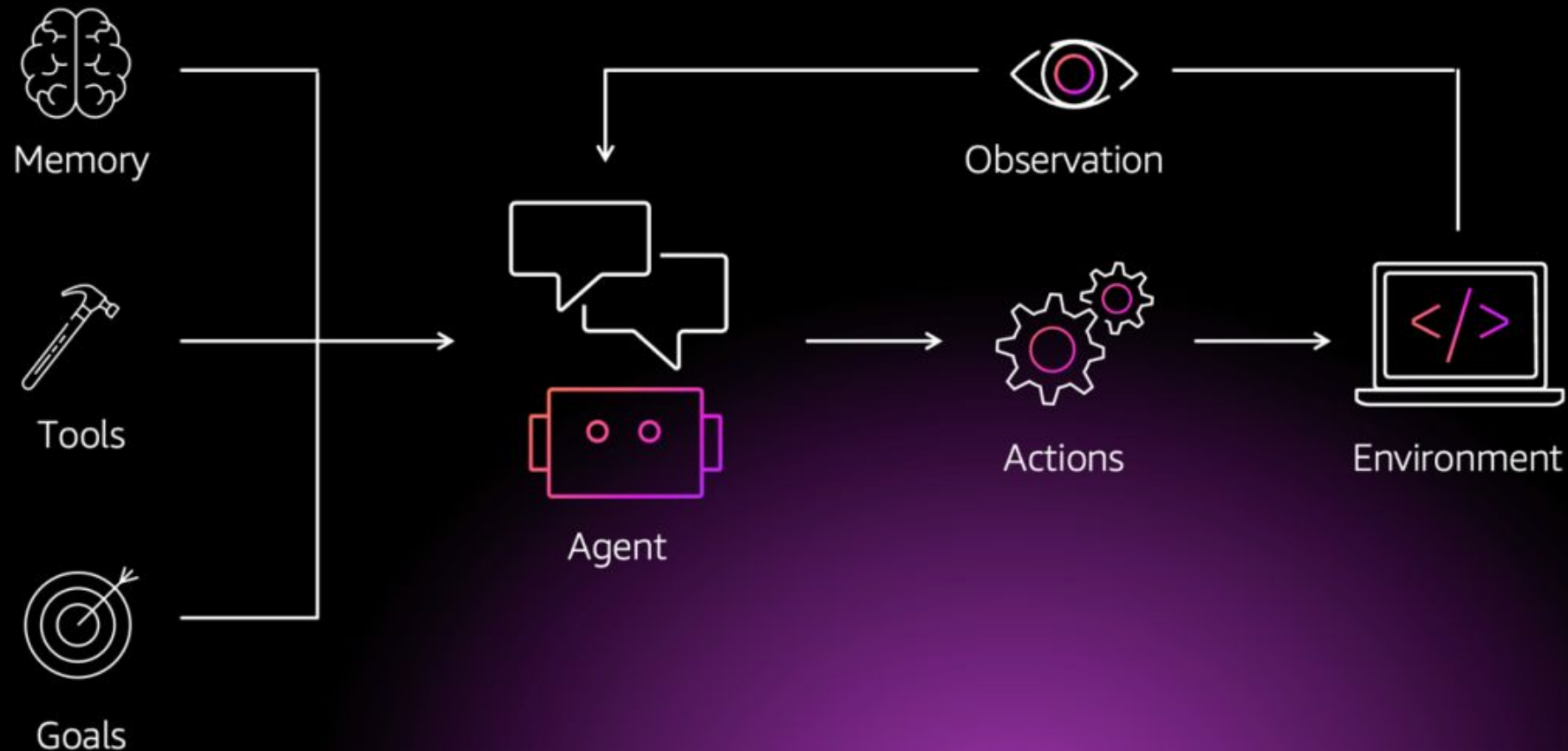
- ▶ AI systems that **perceive**, **reason**, **plan**, and **act** to fulfill goals.
- ▶ **Agent loop:** Observe → Plan → Act → Reflect → Learn
- ▶ Inspired by cognitive science, robotics, and autonomous agents.

Core Components:

- ▶ Memory
- ▶ Planning
- ▶ Tool-use (plugins, APIs)
- ▶ Feedback-driven behavior

What is Agentic AI?

What is agentic AI?



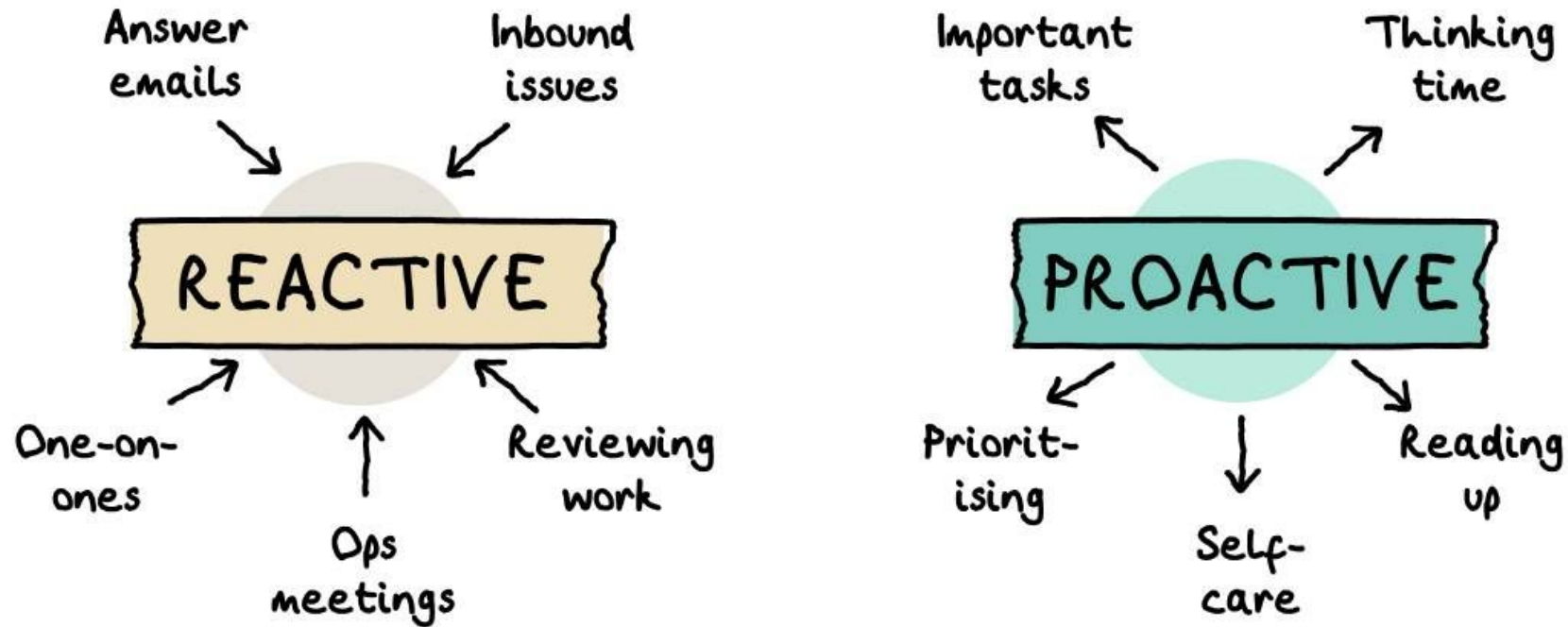
Reactive vs. Proactive Agents

Feature	Reactive	Proactive
Response	Immediate	Goal-oriented
Planning	None	Yes
Learning	Event-triggered	Self-initiated
Example	Chatbots	Auto-GPT, Personal Assistants

Proactive agents generate goals, schedule actions, and evaluate results even without user prompts.

Reactive vs. Proactive Agents

Reactive vs. Proactive Time

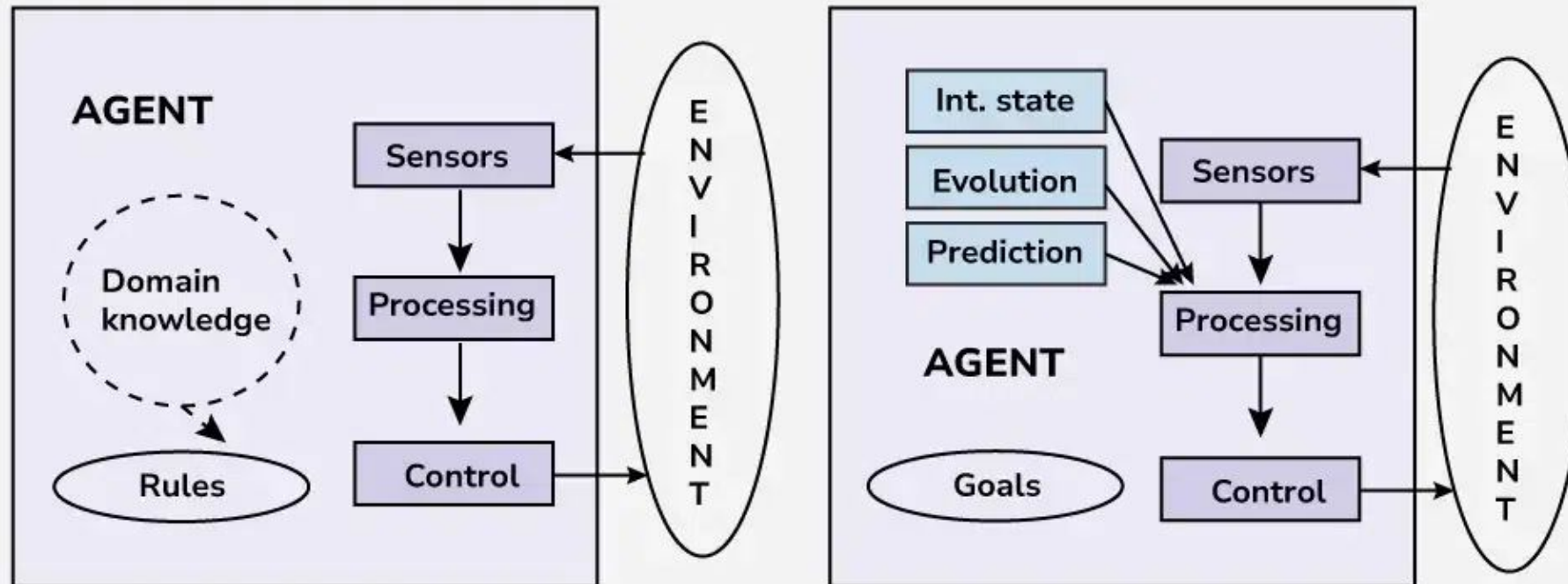


Reactive vs. Proactive Agents

Reactive

VS

Deliberative
agents

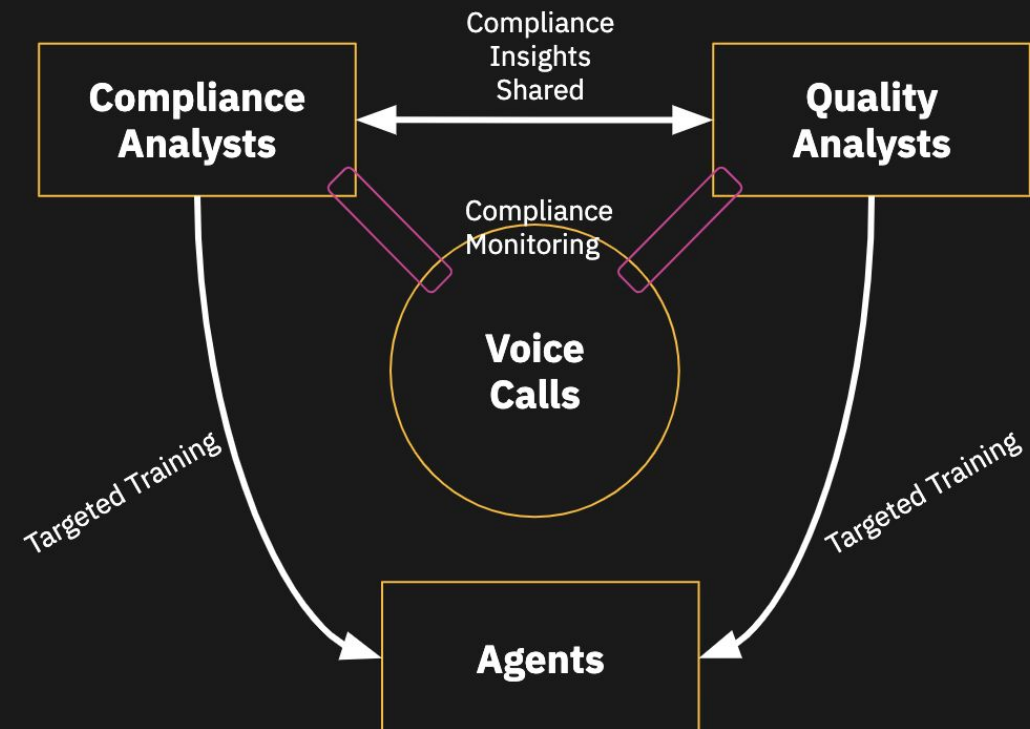


Reactive vs. Proactive Agents

REACTIVE QA



PROACTIVE QA

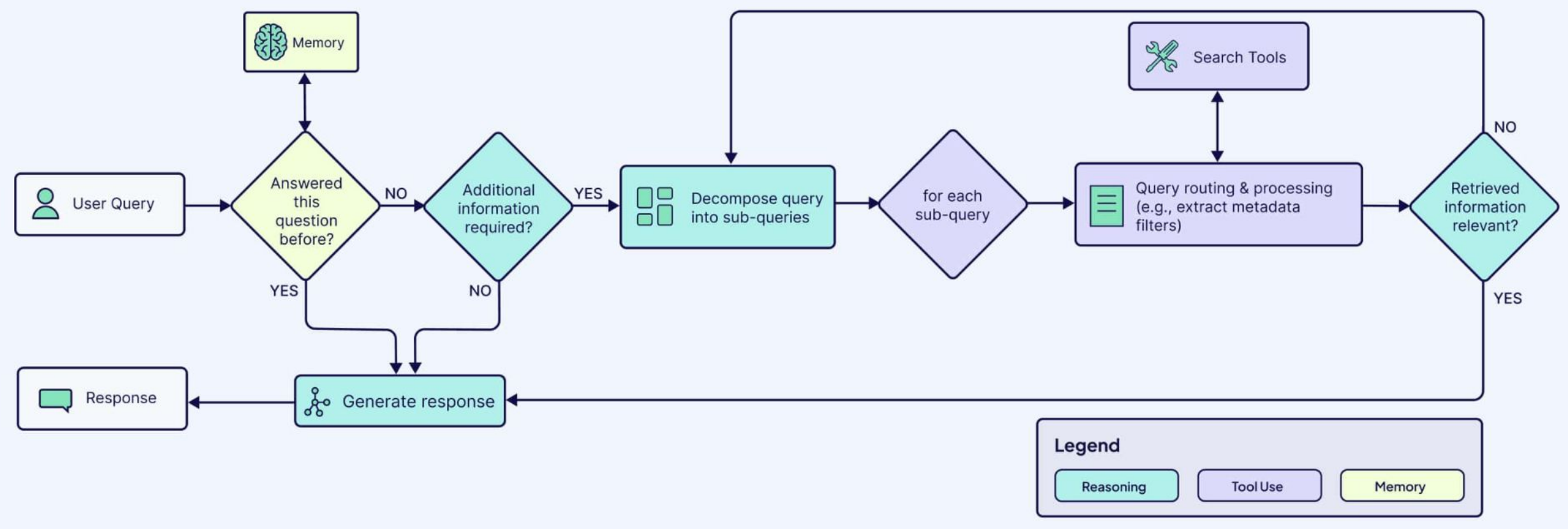


- ▶ **Perceive:** Multimodal input (text, vision, memory)
- ▶ **Interpret:** NLP/vision models
- ▶ **Plan:** Task decomposition (e.g., LangChain chains)
- ▶ **Act:** API/tool calling
- ▶ **Reflect & Learn:** Refine memory or adjust goals

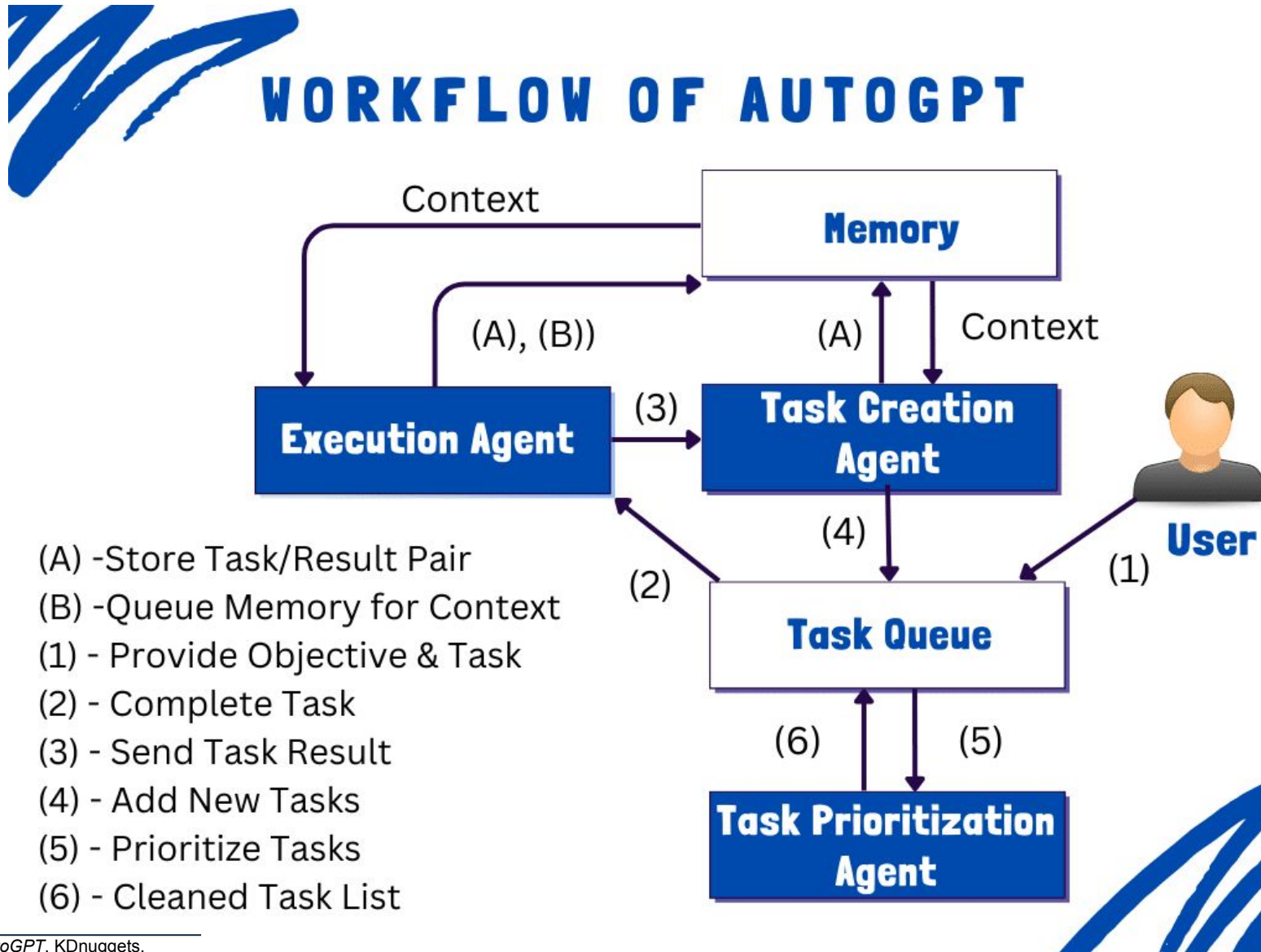
Applications:

- ▶ Personal AI assistants
- ▶ Research co-pilots
- ▶ Automated workflows

Agentic RAG Workflow



Open-Source Agentic Frameworks



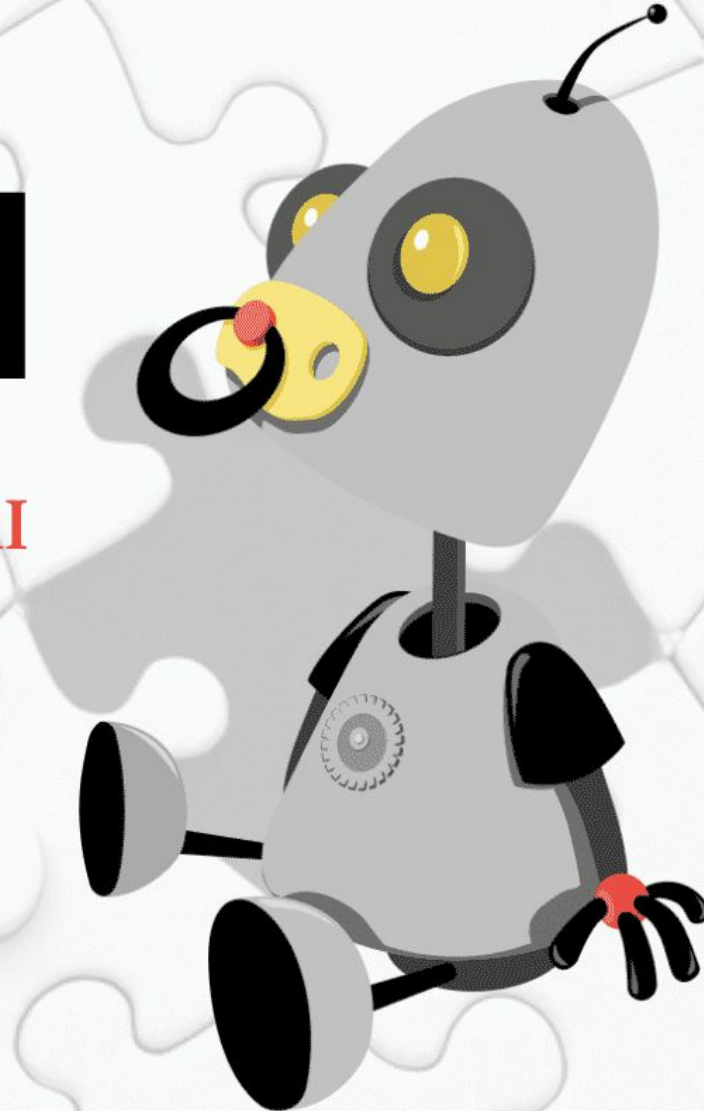
- ▶ **Open-source project** that chains GPT calls to build autonomous agents.
- ▶ Uses **long-term memory**, file storage, and self-generated tasks.
- ▶ **Requirements:**
 - Task input
 - Internet / API access
 - Feedback loop

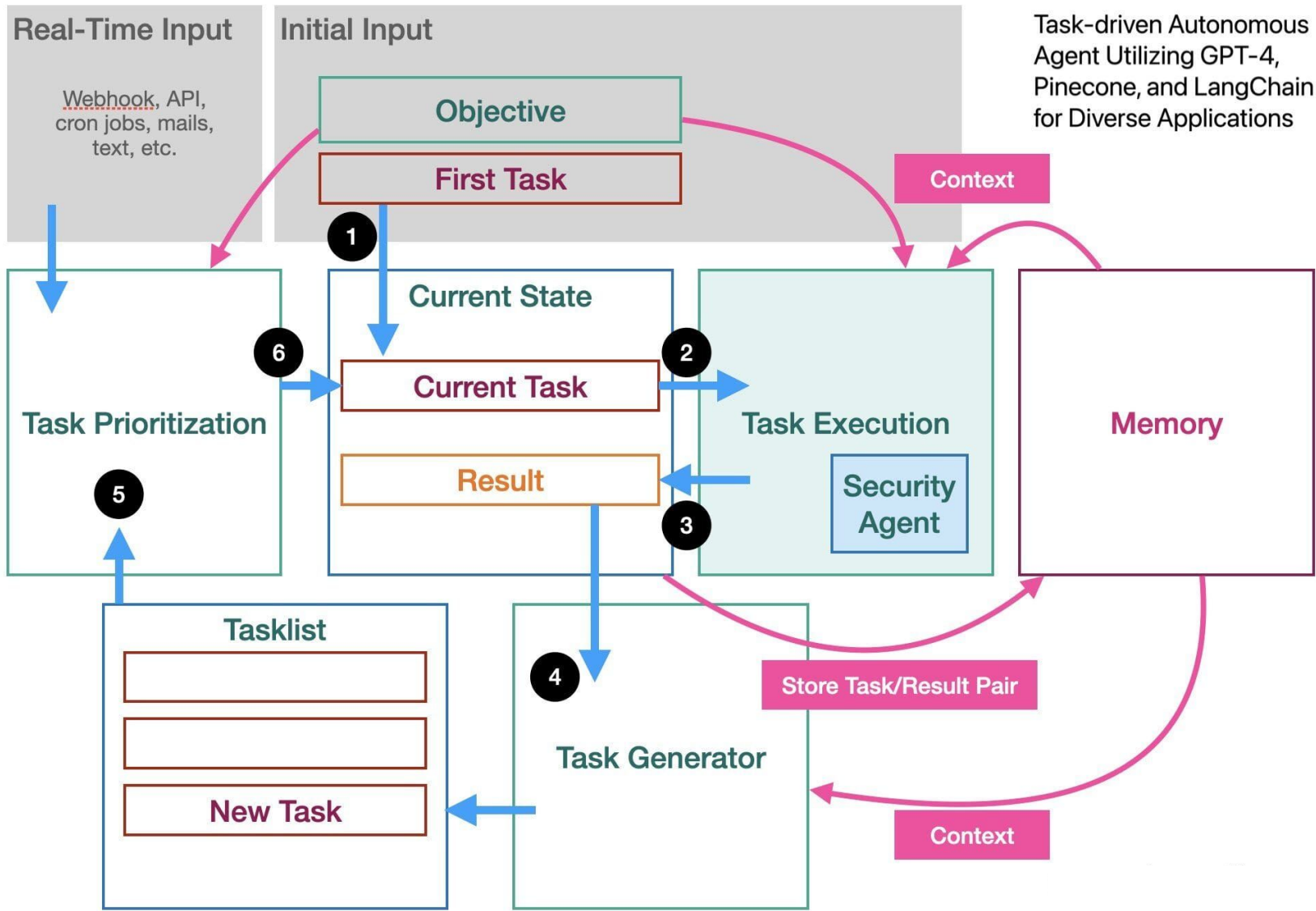
GitHub: <https://github.com/Torantulino/Auto-GPT>



Baby AGI

The Birth of a Fully Autonomous AI

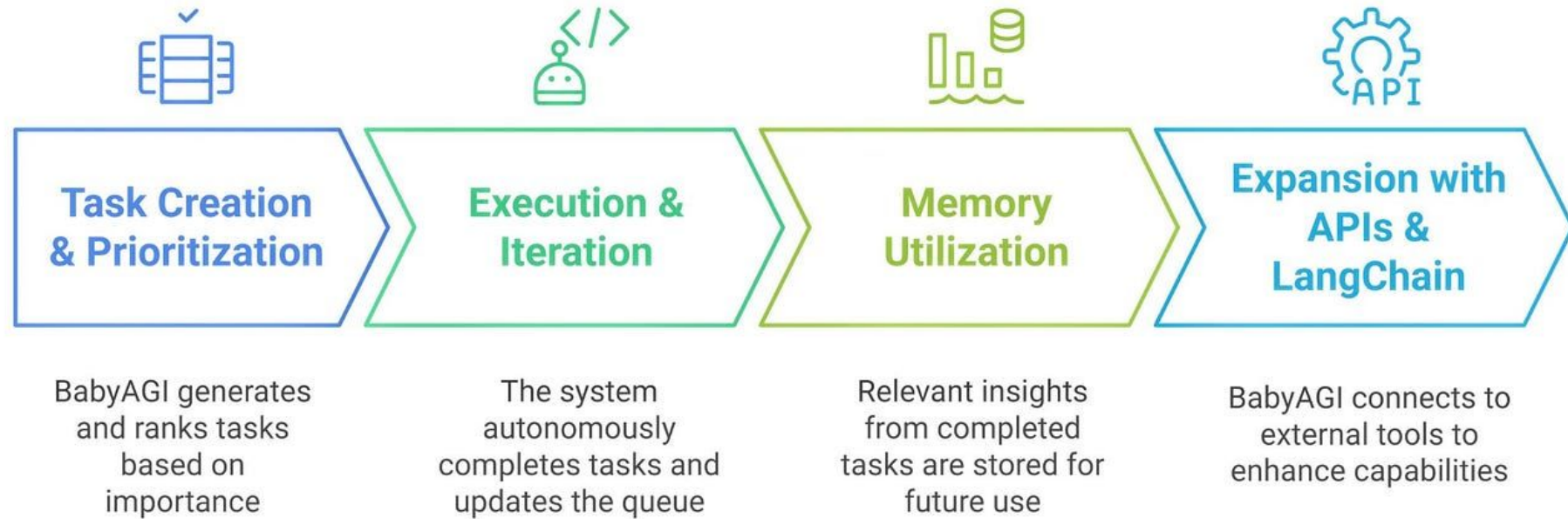




- ▶ **Lightweight Python agent loop**
- ▶ Generates tasks from high-level objectives
- ▶ Runs tasks, stores results, and generates new tasks iteratively
- ▶ Emphasizes simplicity and minimalism

GitHub: <https://github.com/yoheinakajima/babyagi>

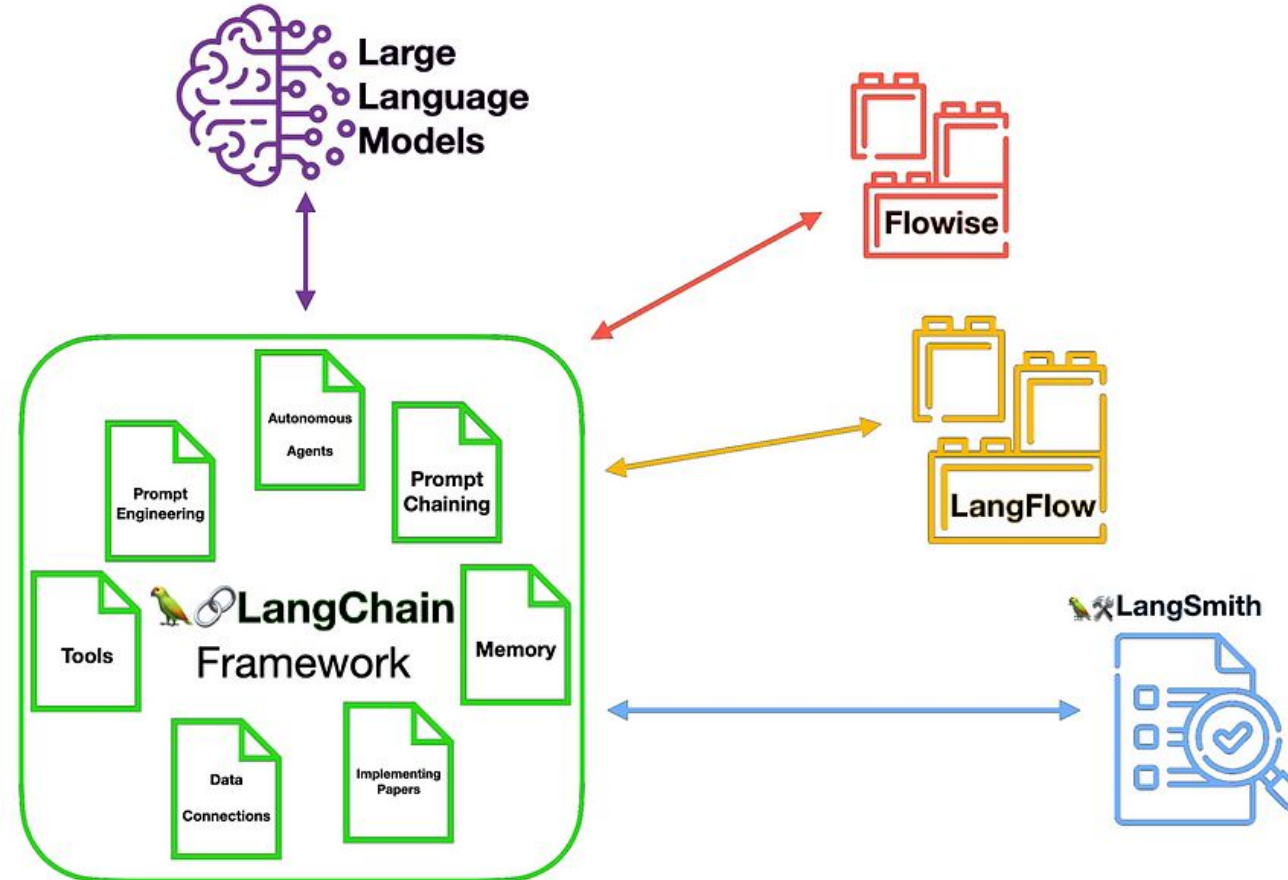
How BabyAGI Works:-

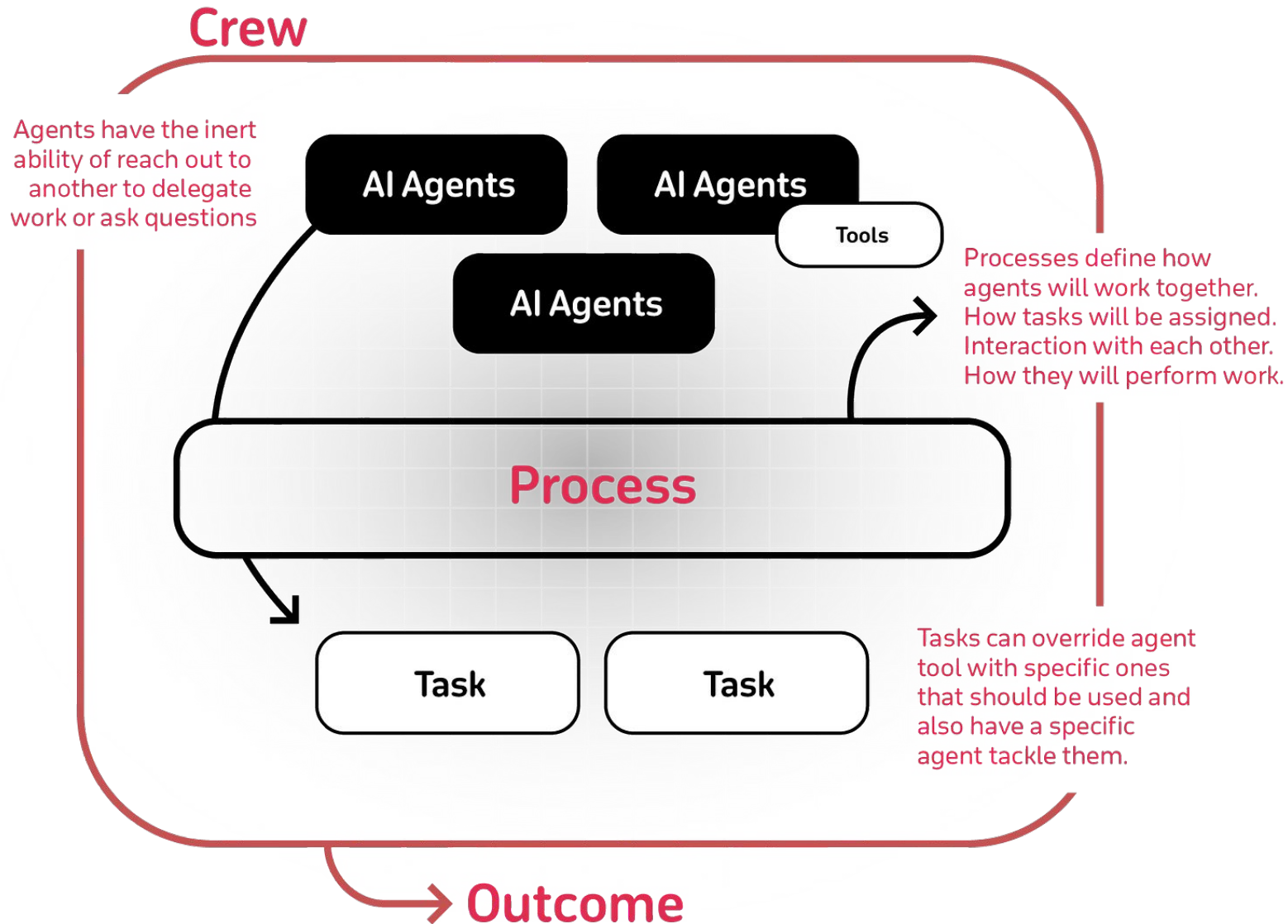


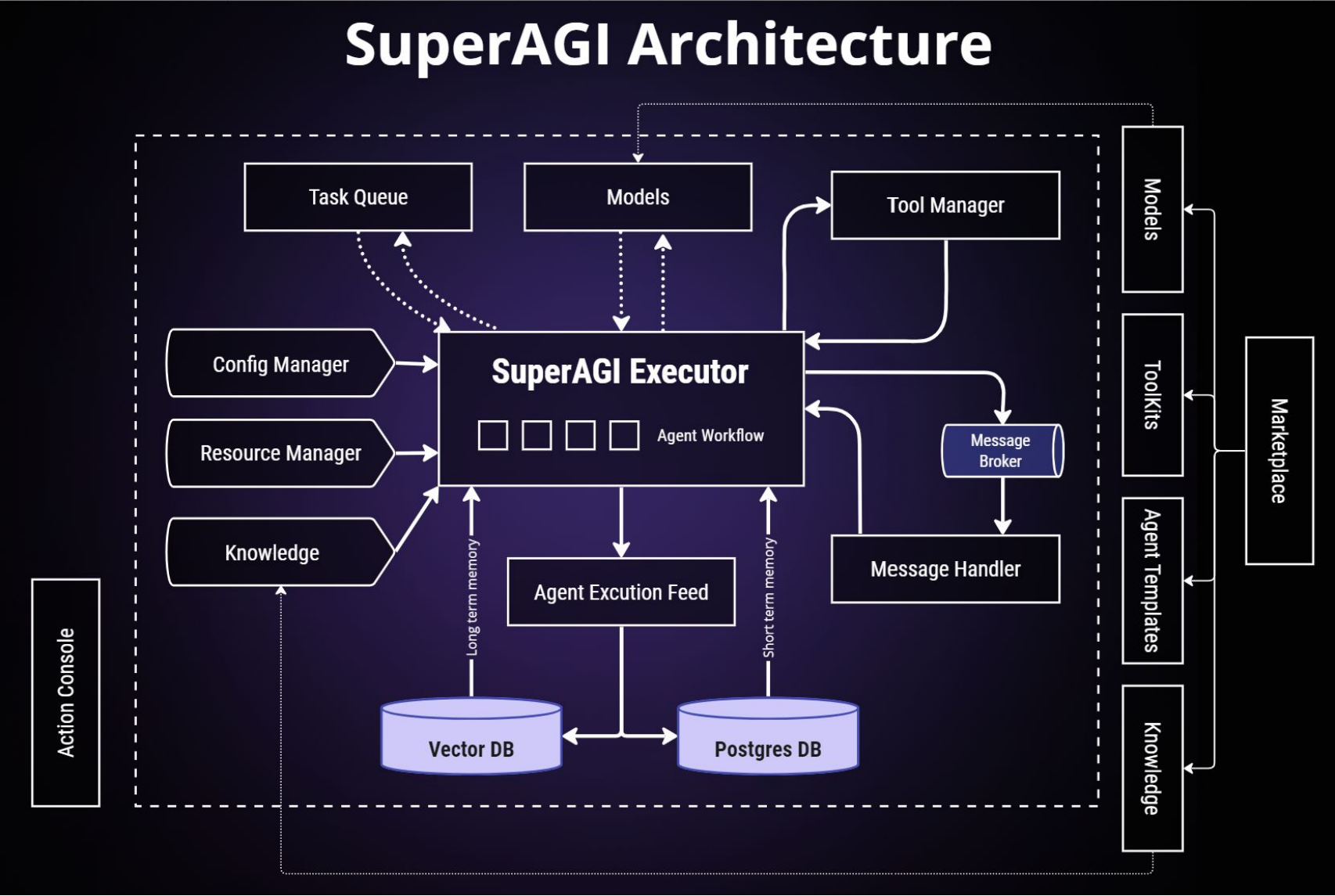
- ▶ **LangChain:** Framework for chaining LLMs with tools, APIs, and memory. Enables complex workflows and agentic behavior.
- ▶ **CrewAI:** Multi-agent coordination platform for collaborative task execution.
- ▶ **SuperAGI:** Production-grade agent platform supporting extensibility, monitoring, and deployment.
- ▶ **OpenDevin:** Developer-oriented agentic IDE for automating software engineering tasks.



LangChain Ecosystem

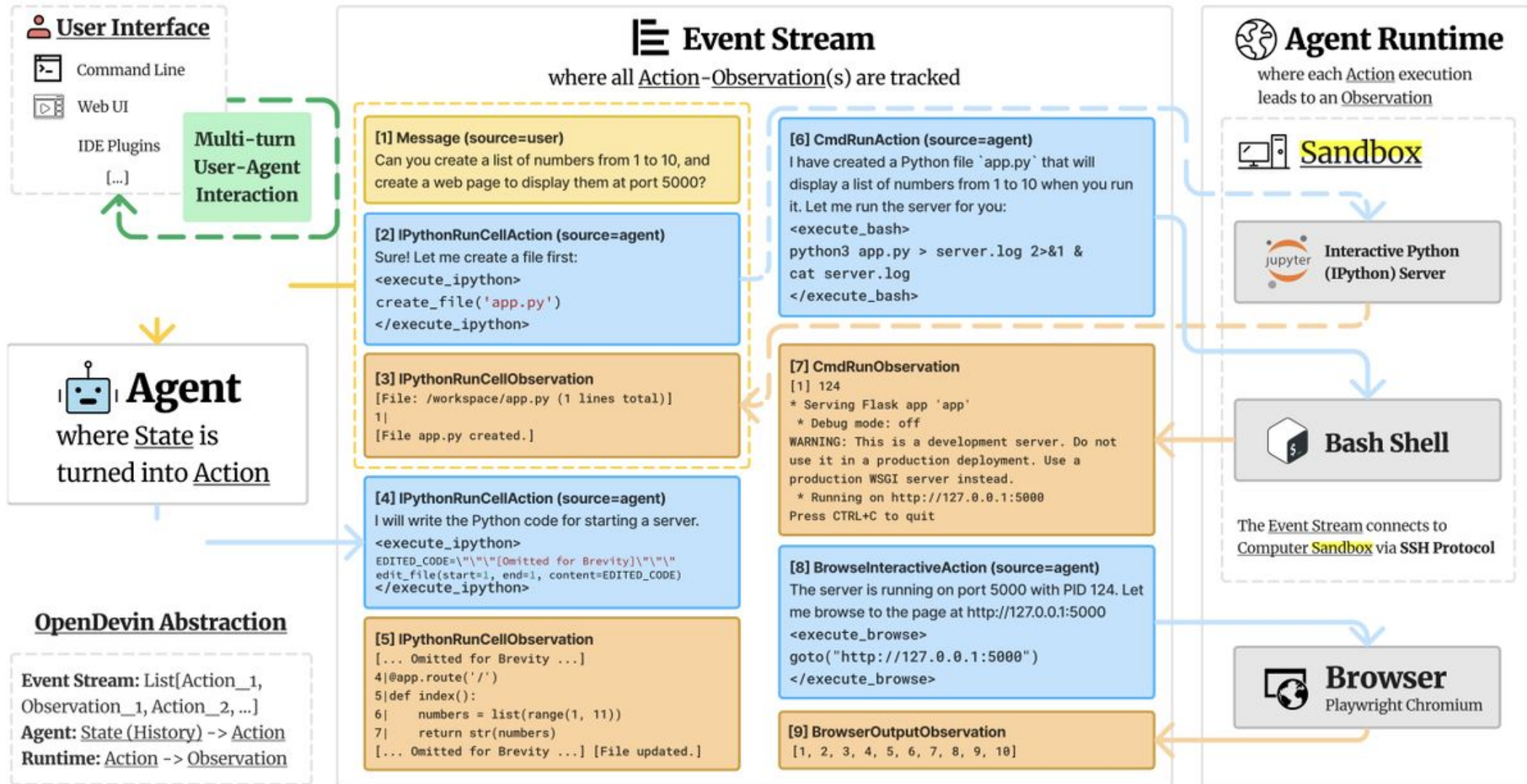






SuperAGI Core Features & Capabilities





Continual Learning in Agents



Continual Learning Machine Learning Model

Data Stream



Up-To-Date
Data Application



Why Continual Learning?

- ▶ Agentic AI should evolve with time:
 - Learn from feedback
 - Adapt to new tasks
 - Retain useful knowledge

Real-world scenarios:

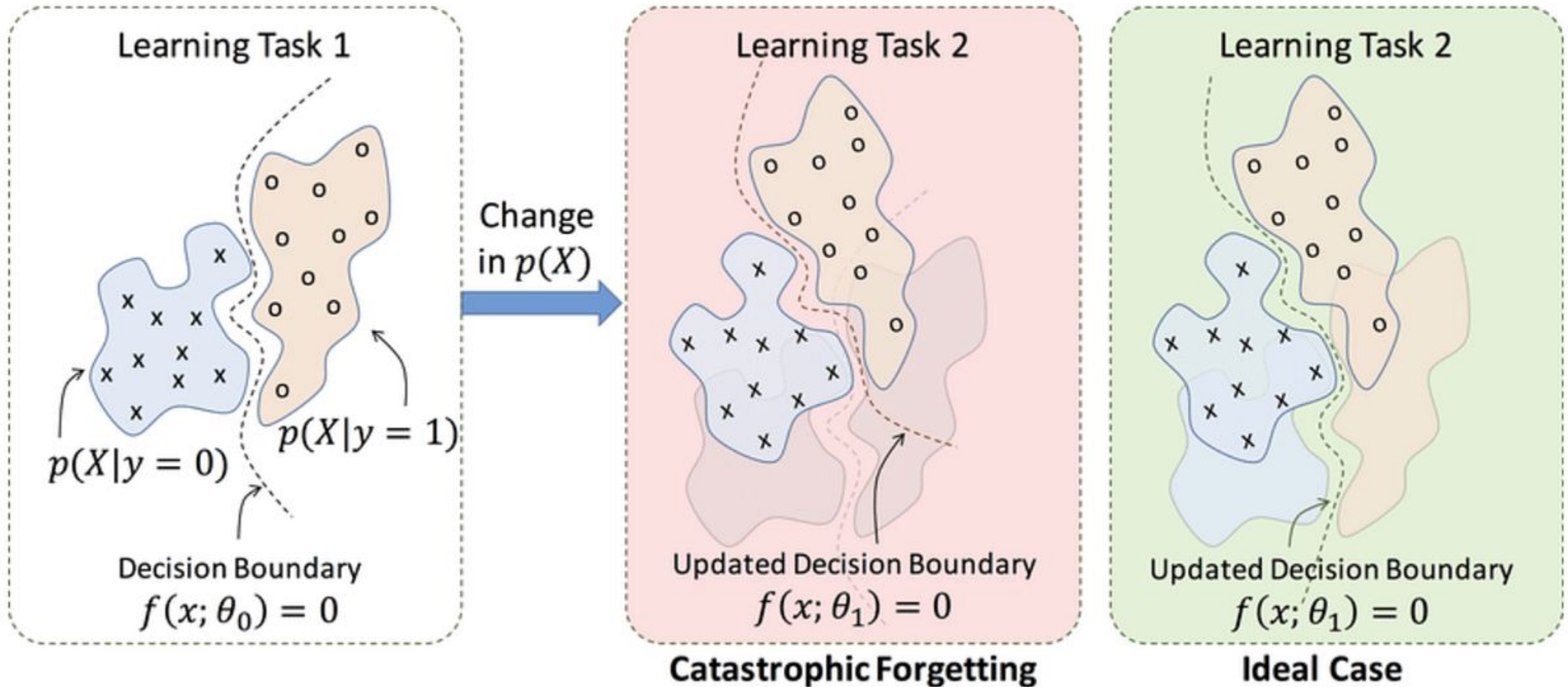
- ▶ Personalization
- ▶ Environment drift
- ▶ Avoiding forgetting

- ▶ **Catastrophic forgetting:** Agents may lose previously acquired knowledge when learning new tasks.
- ▶ **Stability-plasticity dilemma:** Balancing the ability to learn new information (plasticity) with retaining old knowledge (stability).
- ▶ **Modality drift:** Changes or shifts in the types or distributions of input modalities over time.
- ▶ **Scaling memory:** Managing both semantic (general knowledge) and episodic (specific experiences) memory as agents interact with the world.

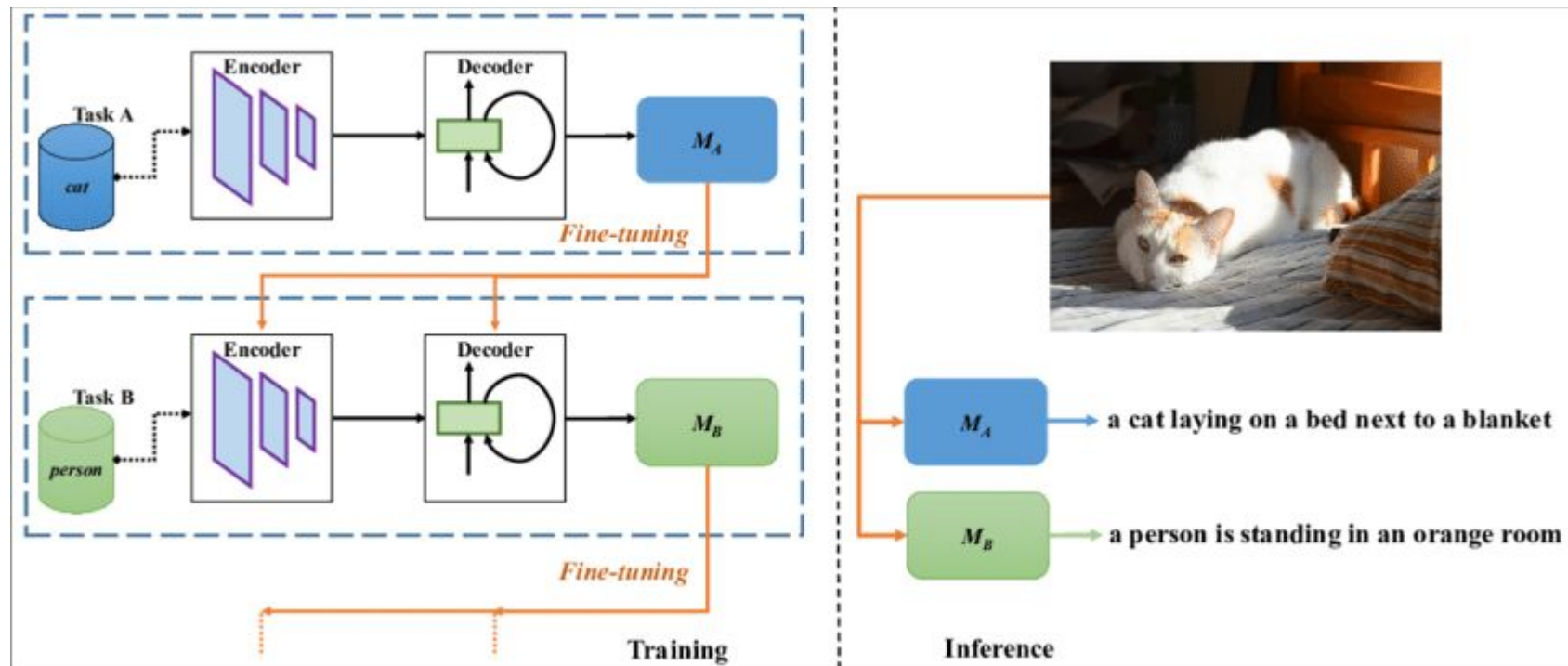
Solutions:

- ▶ Replay memory (experience replay)
- ▶ Progressive networks
- ▶ Elastic weight consolidation (EWC)

Catastrophic forgetting



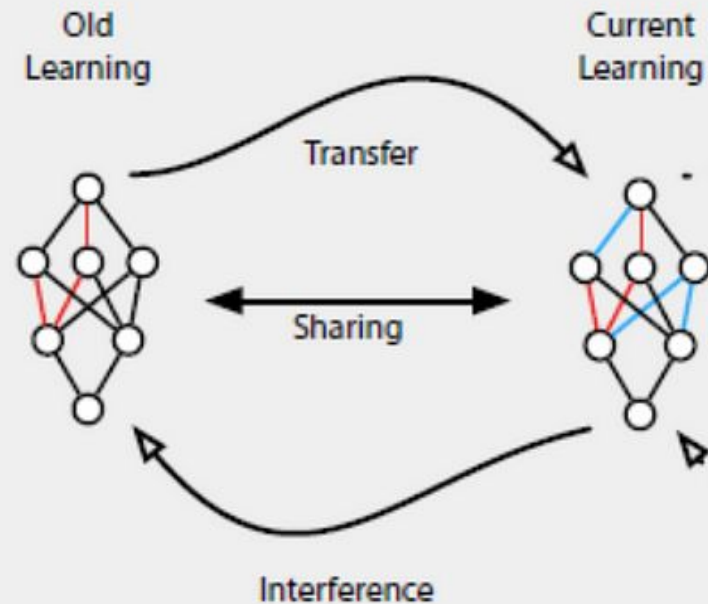
Catastrophic forgetting



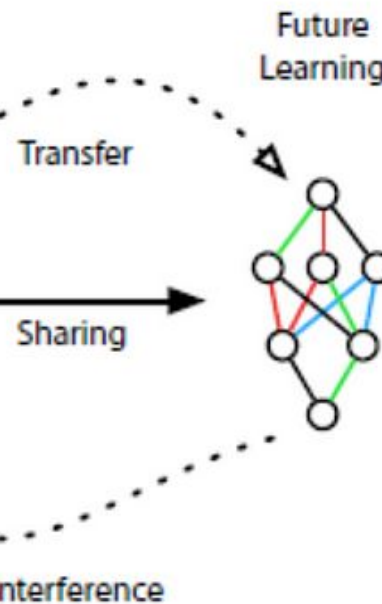
Stability-plasticity dilemma

A. Transfer – Interference Trade-off

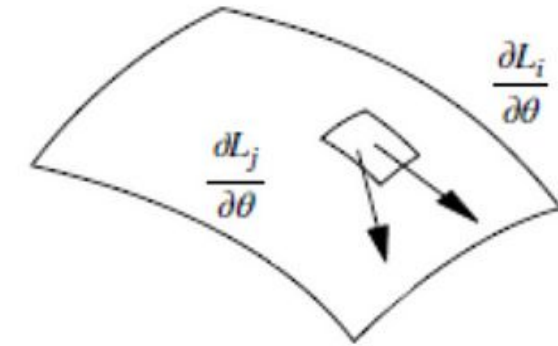
Stability – Plasticity Dilemma



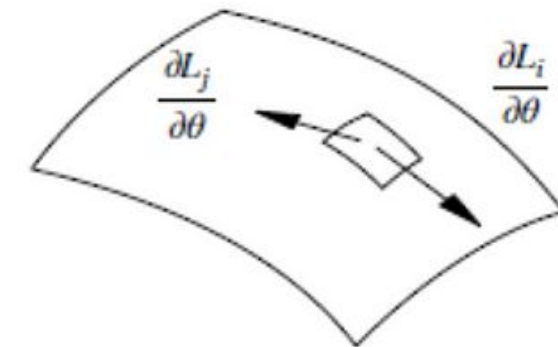
Stability – Plasticity Dilemma



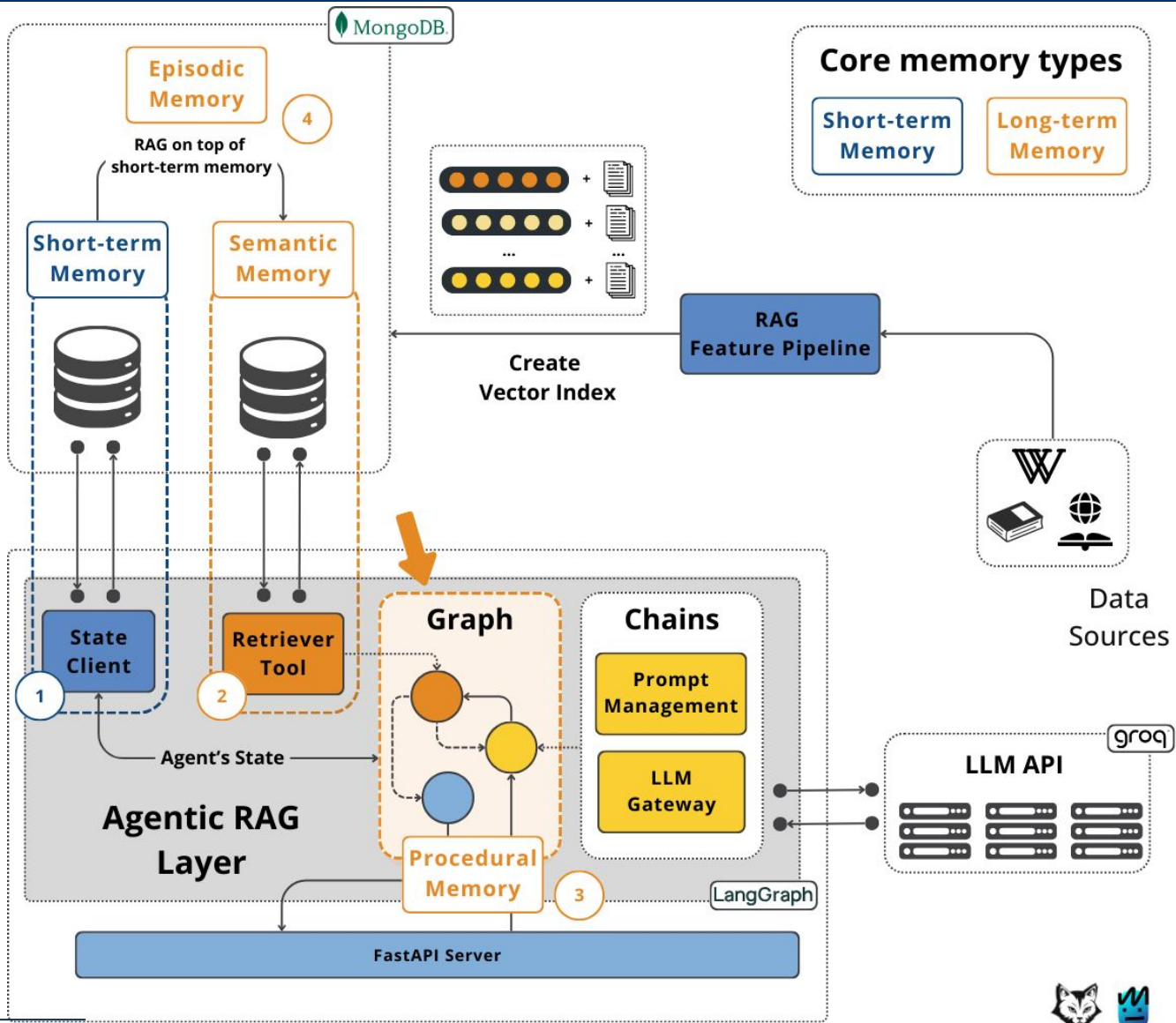
B. Transfer



C. Interference



Scaling memory



Safety, Ethics Control

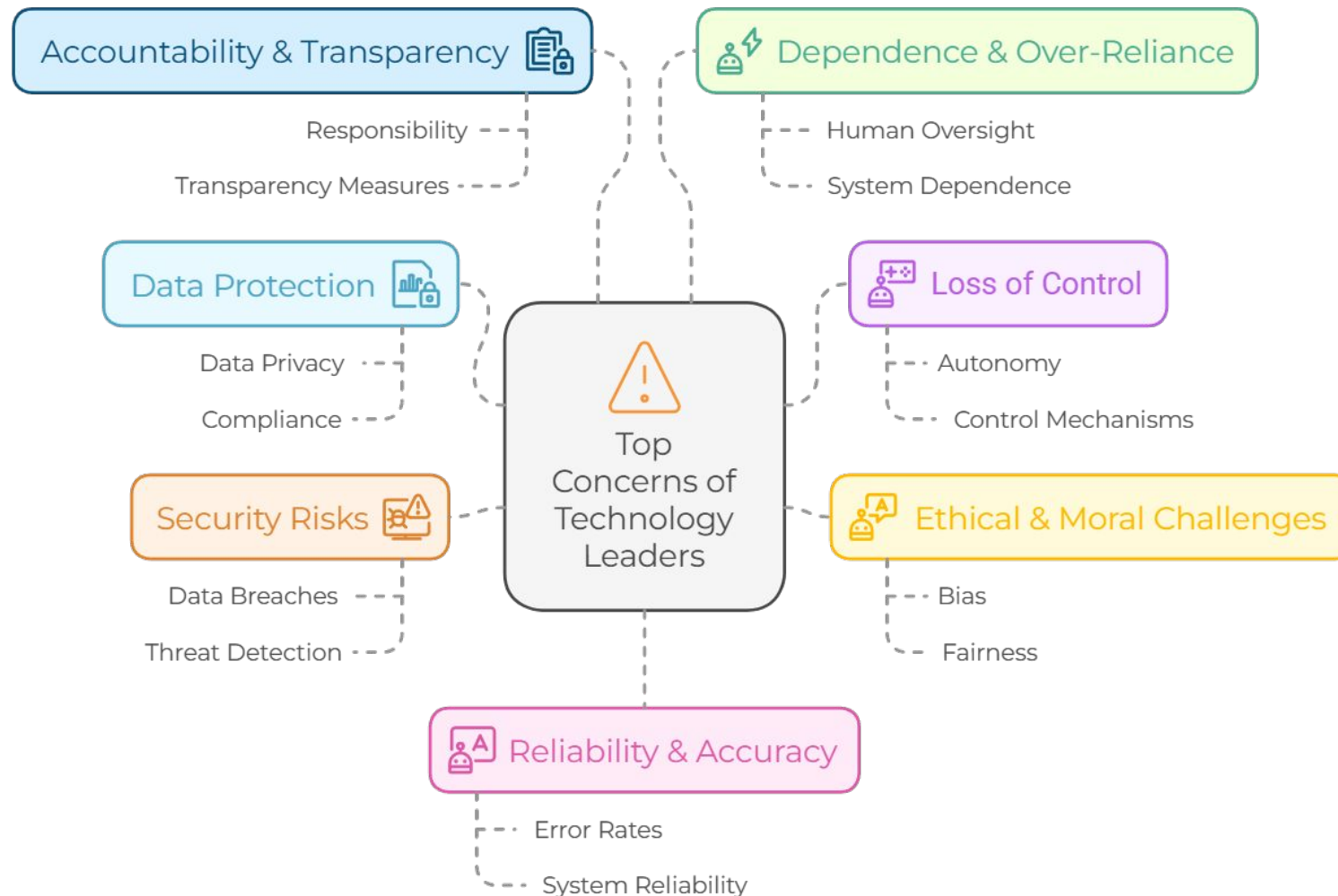
► Risks:

- Autonomous actions without human oversight
- Errors in tool or API use
- Prompt injection or goal hijacking
- Recursive self-improvement without constraints

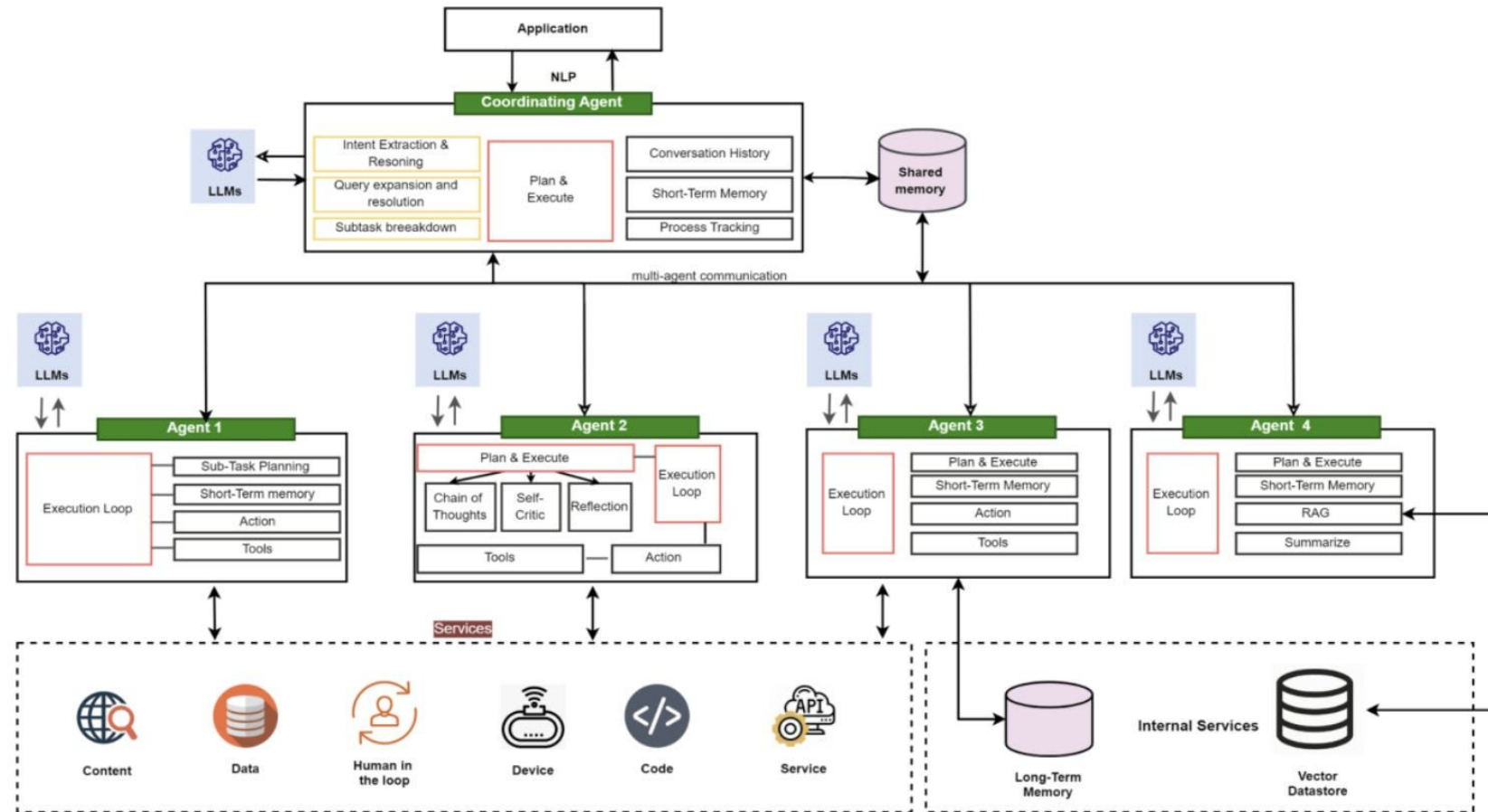
Mitigations:

- Approval-based execution (human-in-the-loop)
- Guardrails and policy enforcement
- Explainability (XAI) for agent decisions

Safety in Agentic Systems



Safety in Agentic Systems



The diagram depicts an *example of* multi-agent architecture of specialized agent functionality. Specialized functionality is a form of agentic patterns and could be exhibited by any agent depending on the use case.



- ▶ **Alignment:** Are agents acting in users' interests?
- ▶ **Accountability:** Who is responsible for AI actions?
- ▶ **Control:** Can humans override or audit decisions?

Key Areas:

- ▶ Fairness in multimodal data
- ▶ Bias amplification (vision + language)
- ▶ Transparency in decision-making

Future Directions



- ▶ **Multimodal grounding in real-world environments:** How can agents robustly connect language, vision, and action in dynamic, noisy settings?
- ▶ **World model integration:** Combining temporal and spatial memory for persistent, context-aware reasoning.
- ▶ **Agent collaboration:** Enabling effective communication and coordination in multi-agent systems.
- ▶ **Scaling memory-efficient agents:** Designing architectures that scale to long-term, large-scale memory without prohibitive costs.
- ▶ **Robust reward models:** Developing reliable reward and feedback mechanisms for open-ended, real-world tasks.
- ▶ **Meta-cognition:** Building agents that can introspect, self-monitor, and adapt their own reasoning processes.

- ▶ Multimodal models (CLIP, VisualBERT, FLAVA) integrate perception and language for richer AI understanding.
- ▶ Agentic AI enables proactive, goal-driven, and autonomous behavior.
- ▶ Open-source frameworks (Auto-GPT, BabyAGI) are accelerating research and applications.
- ▶ Ethical, safe, and explainable deployment is essential for real-world impact.
- ▶ The future of AI lies in continual learning and intelligent autonomy.

- [1] Radford, A., Kim, J. W., Hallacy, C., et al. (2021).
CLIP: Learning Transferable Visual Models From Natural Language Supervision.
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<https://openai.com/blog/>

Credits

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This project benefited from external collaboration, and we acknowledge their contribution with gratitude.